



ARCHITECTURE RESEARCH REPORT

AMD AI INFRASTRUCTURE

A decade of architectural evolution

EPYC • Instinct • Pensando • Adaptive computing

2015-2026

From CPU and GPU generations to programmable networking and complete AI systems.

Product histories for EPYC, Instinct, Pensando DPUs and AI NICs, with adaptive accelerators, rack platforms and software. Built from Hot Chips, ISSCC, ISCA, MICRO, HPCA, ASPLOS and networking research.

Framework edition: normalized comparisons, a claims ledger, four system-era diagrams, conference index and codename crosswalk.

Coverage and research cutoff: 27 September 2026

Product generations first • Sources collected at the end

Reading guide

Chapters 1-7 follow the product families. Chapter 8 preserves the fixed comparison schemas; Chapters 9-10 connect the stack and its engineering trade-offs. Chapters 11-13 provide the conference index, codename crosswalk and sources.

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Scope

Research cutoff: **27 September 2026**. Conference record through Hot Chips 2026, ISSCC 2026 and Advancing AI 2026; ROCm announcement coverage through August 2026. ISSCC 2027 has not yet occurred.

Families and weights: deep coverage of EPYC and Instinct; full generation-level coverage of scale-up links, Pensando DPUs and AI NICs; proportional coverage of adaptive products, systems, software and client derivatives.

Start year: 2015, with earlier silicon only where it explains the first covered generation.

Cutoff: 27 September 2026; product status and roadmap dates are frozen to this date.

Audience and focus: experienced CPU and datacenter architects; architectural changes, physical implementation and their system consequences.

Boundaries: include acquired ancestry; AMD has no separately established merchant datacenter switch-ASIC or optical-transceiver line in this reviewed portfolio, so partner switches and optics appear at the system layer. This is a product-focused disclosure survey, not a census of every AMD-affiliated paper or SKU.

Executive synthesis

AMD's server history is a sequence of changes to the boundary of a processor. Naples assembled four complete CPU SoCs. Rome moved memory and I/O into a central die and replicated smaller compute chiplets. Milan enlarged the cache-sharing domain; V-Cache added capacity vertically. Genoa and Turin then offered different combinations of core density, cache and memory bandwidth around related platforms. Venice extends the platform toward sixteen memory channels and a much larger accelerator-host role.

Instinct followed a parallel but distinct path: graphics-derived compute, a dedicated CDNA matrix architecture, two GPU dies, stacked compute chiplets, and finally a switched rack-scale accelerator domain. The relevant unit of comparison changed along the way. MI250X is one package containing two logical GPU devices. MI300X can expose eight XCDs as one GPU. Helios connects seventy-two GPUs, but its quoted rack HBM remains distributed memory rather than one uniformly local store.

Pensando fills two different requirements. DPUs place networking, security and storage services outside tenant CPU execution. AI NICs move accelerator traffic across a scale-out network. Pollara and Vulcano complement the GPU scale-up fabric; Salina addresses infrastructure services, including access to retained inference state. Xilinx adds programmable spatial processing and networking, while system design and ROCm turn component capabilities into deployable platforms.

The headline figures show that more arithmetic does not automatically create a balanced machine:

Era / flagship pairing	CPU maximum cores; DRAM GB/s/socket	GPU dense BF16 / FP8 TFLOP/s	GPU memory capacity; TB/s	Scale-up domain
2017: Naples / MI25	32; 170.7	n/a / n/a	16 GB HBM2; 0.484	PCIe system-dependent
2019-2020: Rome / MI100	64; 204.8	92.3 / n/a	32 GB HBM2; 1.229	4-GPU IF hive
2021-2022: Milan / MI250X	64; 204.8	383 / n/a per package	128 GB HBM2e; 3.2	Frontier: 4 packages, 8 GPU devices
2023: Genoa / MI300X	96; 460.8	1307.4 / 2614.9	192 GB HBM3; 5.3	8 GPUs
2024: Turin / MI325X	128 or 192; 614.4 current	1307.4 / 2614.9	256 GB HBM3e; 6.0	8 GPUs
2025: Turin / MI355X	128 or 192; 614.4 current	2517 / 5033	288 GB HBM3e; 8.0	8 GPUs per baseboard
2026: Venice / MI455X	256; 1024 RDIMM / 1638.4 MRDIMM	5033 / 20133	432 GB HBM4; 23.3	72 GPUs in Helios

These are era pairings and representative maxima, not mandatory node bills of materials. MI455X and Venice have launch and production disclosures; broad general availability is not established by those disclosures alone. The 2026 rows describe the published platform while customer rollout proceeds. Older rows describe historically available products, without asserting that every legacy card remains orderable.

Three resource balances are particularly revealing. At current DDR5-6400 specifications, Turin offers 4.8 GB/s per core at 128 cores and 3.2 GB/s per core at 192 cores. MI355X offers roughly 0.00318 HBM bytes per dense BF16 FLOP, compared with 0.00405 for MI300X. MI455X reverses that BF16 trend to about 0.00463, but its much larger FP8 rate gives only about 0.00116 HBM bytes per FP8 FLOP. Reuse, layout and precision selection therefore remain central even after a large HBM upgrade.

The other decisive change is specialization. MI455X's published FP64 peak is 5 TFLOP/s, compared with 78.6 TFLOP/s for MI355X; the separate MI430X branch is announced for HPC in 2027. Meanwhile MI350P places CDNA 4 in a conventional PCIe card, and the dense EPYC branches choose different cache and memory resources per core. A useful comparison must follow the workload and deployment branch, not only the generation number.

The engineering direction is clear: reusable compute dies, increasingly specialized infrastructure dies, deeper integration near memory, and a wider system boundary. The costs are equally concrete: additional locality boundaries, more complex partitioning, greater package and thermal demands, and a stronger dependency on communication software. The 2027-2030 roadmap extends this direction, but its names and intended dates should not be mistaken for completed implementations.

How to read the comparisons

All compute rates are vendor peak **dense** rates unless marked otherwise. A multiply-add follows the vendor's two-operation convention. Per-package, per-card and per-rack quantities are identified. FP8 and microscaling formats are named; sparse figures are not substituted for dense figures. These rates are not application benchmarks.

Link figures use decimal GB/s and state direction. Gb/s is divided by eight; a quoted bidirectional total is divided by two before comparison with a per-direction figure. HBM bandwidth is compared with one direction of the link. PCIe encoding ceilings exclude transaction overhead, while Ethernet line rates exclude protocol overhead. TDP, cTDP, TBP and the newer default CPU-power terminology retain their vendor definitions.

"n/d" means not established in the reviewed primary disclosure, rather than proof that no document exists; "n/a" means not applicable. The fixed schemas are preserved in the wide comparison tables and companion CSVs. Their rows distinguish shipping, limited availability, announced, roadmap and repositioned products. A separate chronology records first material disclosure and commercial availability. Material document conflicts are called out once, beside the affected product.

1. EPYC products and technology evolution

1.1 The product sequence

Product family	Introduction	Core architecture and maximum cores	Principal change
EPYC 7001, Naples	2017	Zen; 32	Four reusable Zeppelin SoCs in one server package
EPYC 7002, Rome	2019	Zen 2; 64	Up to eight compute chiplets around a central I/O die
EPYC 7003, Milan	2021	Zen 3; 64	Eight cores share one 32 MB L3 within each compute chiplet
EPYC 7003X, Milan-X	2022	Zen 3; 64	3D V-Cache raises maximum package L3 to 768 MB
EPYC 9004, Genoa	2022	Zen 4; 96	SP5, DDR5, PCIe 5.0, CXL and AVX-512
EPYC 9004X, Genoa-X	2023	Zen 4; 96	Up to 1,152 MB L3 for technical computing
EPYC 97x4, Bergamo	2023	Zen 4c; 128	Denser cores and cache organization for cloud throughput
EPYC 8004, Siena	2023	Zen 4c; 64	Smaller SP6 platform for edge, telco and constrained servers
EPYC 9005, Turin	2024	Zen 5; 128, or Zen 5c; 192	Wider execution, denser compute options and faster DDR5
EPYC 4004 / 4005	2024 / 2025	Zen 4 / Zen 5; up to 16	AM5 entry server branch with close desktop ancestry
EPYC 8005, Sorano	2026	Zen 5; 84	SP6 edge and telco branch with six DDR5-6400 channels
EPYC 9006, Venice	2026	Zen 6 / Zen 6c; up to 256	New SP7/SP8 portfolio; flagship SP7 adds 16 memory channels and PCIe 6.0

The main server lineage is Naples → Rome → Milan → Genoa → Turin → Venice. X variants, dense-core variants and smaller-platform variants are branches of that lineage. The numbering does not imply that every product with a later launch date introduces a new CPU core.

1.2 Naples: EPYC 7001, 2017

Naples brought AMD back into mainstream high-performance x86 servers with up to 32 cores, eight DDR4 channels and 128 PCIe 3.0 lanes. The package contained four 14 nm Zeppelin dies. Each die was a complete SoC with eight Zen cores, memory controllers, I/O and Infinity Fabric connectivity.

This was a product-family strategy from the beginning. Zeppelin could serve desktop Ryzen, high-end desktop Threadripper and EPYC by changing how many dies were packaged and how their interfaces were configured. Sharing a substantial silicon building block spread development cost across markets and avoided making a single very large server-only die.

The cost of this reuse appeared in the memory topology. Each die owned two memory channels. A core accessing another die's memory crossed the on-package fabric, so one socket contained meaningful locality boundaries. Operating-system placement, memory allocation and thread affinity mattered even before a workload crossed into a second socket. Naples should therefore be understood as a coherent multi-die server, rather than a monolithic 32-core processor with eight equally local channels.

The Zeppelin ISSCC presentation makes the difference concrete: in its illustrated configuration, local memory access was approximately 90 ns, another die's memory within the socket approximately 145 ns, and a single-hop access to the other socket approximately 200 ns. These are measurements for that configuration, but they expose the architectural cost of crossing each boundary.

Within a Zeppelin die, two four-core complexes each shared an 8 MB L3. The local cache-sharing unit was therefore smaller than the eight-core die. Distinguishing the core complex, die and socket is essential to understanding the changes that followed.

What the conferences add. Michael Clark's Zen presentation at Hot Chips 2016 introduced the core architecture. Kevin Lepak's EPYC presentation at Hot Chips 2017 explained the enterprise package and system organization. ISSCC 2017 covered the Zen implementation; ISSCC 2018's Zeppelin paper explained how one SoC supported several markets. Together, these sources show that scalability was designed into the die's interfaces and physical implementation, not added only at the packaging stage.

The important inheritance was the combination of a reusable core complex, coherent die-to-die links and a product strategy built around multiple dies. Rome retained that method while changing what belonged on each die.

1.3 Rome: EPYC 7002, 2019

Rome doubled the maximum core count to 64 and made the decisive architectural separation between compute and infrastructure. Up to eight 7 nm core complex dies, or CCDs, connected to a central 14 nm I/O die. The I/O die contained the memory controllers and external interfaces. The platform retained eight memory channels, advanced to DDR4-3200 and PCIe 4.0, and continued on the SP3 socket family.

Each eight-core CCD still contained two four-core complexes. Each complex now had 16 MB L3, giving a full CCD 32 MB and a fully populated 64-core package 256 MB. A large package-level cache total did not mean that every core had one equally local 256 MB cache.

Separating the I/O die solved several problems together. The new process could be devoted to replicated CPU logic. Memory and external I/O no longer had to be reproduced inside every compute die. AMD could vary the number of CCDs without changing the basic memory-controller organization. The desktop Ryzen 3000 family reused the Zen 2 compute-chiplet approach with a different I/O die and fewer chiplets.

Centralizing memory access made Rome's topology more regular than Naples, but did not eliminate locality. A load still traversed a CCD-to-I/O connection, and the I/O die had internal organization. NUMA Nodes Per Socket, or NPS, settings and workload placement remained relevant. Independent memory characterization at ISPASS subsequently measured the practical consequences of this hierarchy.

What the conferences add. Hot Chips 2019's Zen 2 disclosure supplies the core changes, including a larger micro-op cache and wider floating-point execution than the original Zen design. ISSCC 2020 presented both the Zen 2 core and the server/desktop chiplet architecture. ISCA 2021's industrial product paper, *Pioneering Chiplet Technology and Design for the AMD EPYC and Ryzen Processor Families*, explains the manufacturing and reuse rationale across Naples and Rome.

The combined lesson is that Rome's scaling was not simply the result of smaller transistors. It depended on choosing a new boundary between compute, memory control and I/O, then making that boundary economical through package and interconnect design.

1.4 Milan and Milan-X: EPYC 7003, 2021-2022

Milan retained the broad Rome platform: up to 64 cores, eight DDR4 channels, PCIe 4.0 and separate compute and I/O dies. Its defining change was inside the CCD. Zen 3 replaced the two four-core complexes with one eight-core complex sharing 32 MB L3.

This enlarged the domain in which threads could share data through the local L3. It also doubled the directly accessible L3 capacity for a core relative to a Zen 2 four-core complex, while leaving the maximum package L3 at 256 MB. The distinction explains why applications could improve substantially even though the headline core count and total cache capacity stayed unchanged.

Zen 3 also expanded execution resources and improved branch prediction and load/store throughput. Hot Chips 2021 explains those architectural changes. ISSCC 2022 connects them to an eight-slice L3 implementation using a bidirectional ring, denser SRAM choices, physical-design work and power management. This is a useful example of implementation enabling architecture: a larger sharing domain needs a practical interconnect, not just more cache bits.

Milan-X added a 64 MB SRAM die above each CCD's existing 32 MB L3. A fully populated package therefore provided 96 MB per CCD and 768 MB total L3. The added cache was part of the CPU's cache hierarchy; software did not manage it as a separate memory device.

The ISSCC 2022 3D V-Cache paper describes hybrid bonding and the high-bandwidth vertical interface. The base CCD was designed to support stacking, with cache control retained on the base die and the stacked die optimized for storage density. Hot Chips 2021's packaging tutorial places that implementation in AMD's broader integration strategy. Computex 2021 had already demonstrated the technology on a Ryzen prototype, linking the server product to the later Ryzen 7 5800X3D.

The implementation paper reports a 41 mm², 7 nm cache die, a 9 micrometre minimum TSV pitch and more than 2 TB/s of L3 bandwidth. It also reports a four-cycle latency increment. The engineering trade is therefore explicit: much more local capacity through a dense vertical interface, with an access-latency cost, rather than a free enlargement of the original cache.

The benefit is strongest when additional L3 captures a workload's reused data and avoids more expensive memory accesses. This is why engineering simulation and other technical-computing workloads were central to Milan-X's positioning. Additional cache cannot substitute for DRAM bandwidth when a workload mostly streams through data with little reuse.

Milan also introduced SEV-SNP in AMD's confidential-computing progression. Memory protection, virtualization support and platform compatibility are part of the server generation even though they are less visible in core-count comparisons.

1.5 Genoa: EPYC 9004, 2022

Genoa was a new platform as well as a new core. It increased the maximum to 96 Zen 4 cores across twelve eight-core CCDs, used 5 nm compute dies with a 6 nm I/O die, and moved to SP5. Twelve DDR5 channels replaced eight DDR4 channels. PCIe 5.0 and CXL extended the platform's connectivity.

The memory change is particularly significant. At the nominal maximum rates used here, eight DDR4-3200 channels provide 204.8 GB/s of theoretical payload bandwidth, while twelve DDR5-4800 channels provide 460.8 GB/s. The bandwidth increase exceeded the increase from 64 to 96 cores, improving the platform's nominal bandwidth per core.

Zen 4 kept the eight-core, 32 MB L3 complex and doubled private L2 from 512 KB to 1 MB per core. It added AVX-512 instructions using 256-bit execution resources. Software gained the instruction set, but an AVX-512 operation did not imply that every relevant execution datapath was physically 512 bits wide. Turin would change that implementation balance again.

Hot Chips 2023 brings the core and EPYC 9004 platform together. ISSCC 2023's Zen 4 paper supplies the circuit and physical-design view. Its high-frequency result belongs to the demonstrated core implementation and client operating points; it should not be read as a 96-core server's all-core frequency. This is a recurring reason to combine conference disclosures with final product specifications.

Genoa's desktop relative is Ryzen 7000. The common Zen 4 foundation makes client papers useful for understanding the core, but server memory channels, I/O, package power and chiplet count must be taken from EPYC-specific material.

1.6 Genoa-X, Bergamo and Siena: three different optimizations, 2023

Genoa-X extended the V-Cache approach to Zen 4. Twelve CCDs with 96 MB L3 each yielded up to 1,152 MB of package L3. Its purpose was to increase the amount of reusable working data retained near the cores while preserving the SP5 platform. Like Milan-X, it is best understood through the interaction of workload locality, cache capacity and memory traffic.

Bergamo optimized for cloud throughput and density. It used Zen 4c cores and reached 128 cores with eight sixteen-core CCDs. A Bergamo CCD contained two eight-core complexes with 16 MB L3 each. Thus, its 128-core package had 256 MB L3, compared with Genoa's 384 MB at 96 cores.

Zen 4c preserved the Zen 4 instruction-set and software foundation while optimizing the implementation for area and a different frequency/power range. ISSCC 2024 explains this distinction at the physical-design level. Dense cores are not simply ordinary cores placed closer together, nor should the shared ISA be interpreted as identical performance under every workload. Cache capacity, clock frequency, memory contention and placement all influence the resulting behavior.

Siena, sold as EPYC 8004, brought Zen 4c to a smaller single-socket SP6 platform with up to 64 cores, six DDR5 channels and 96 PCIe 5.0 lanes. It targeted edge, telecommunications and deployments constrained by power, physical size or platform cost. Reducing the surrounding platform was as important as changing the core implementation.

AMD's June 2023 Data Center and AI Technology Premiere introduced Bergamo and Genoa-X; Siena followed in September. Hot Chips 2023 supplies the architectural framework for understanding this family. MWC presentations then show why Siena's attributes matter in telecommunications systems.

These products mark a change in how to read the EPYC roadmap: the question becomes which balance of resources fits the workload. Maximum core count, maximum cache capacity and the smallest practical system are three different objectives.

1.7 Turin: EPYC 9005, 2024

Turin extended SP5 with two major compute options: up to 128 Zen 5 cores using sixteen eight-core CCDs, or up to 192 Zen 5c cores using twelve sixteen-core CCDs. The mainstream Zen 5 implementation uses 4 nm compute technology; the dense implementation uses 3 nm. Both connect to the server I/O infrastructure and support twelve DDR5 channels, listed at DDR5-6000 in the April 2025 architecture guide and now at DDR5-6400 on the current 9755 and 9965 product pages. The current comparison therefore uses 614.4 GB/s per socket; the older 576 GB/s figure describes the earlier published configuration.

The maximum cache organizations differ. An eight-core Zen 5 CCD has 32 MB L3, yielding 512 MB across sixteen CCDs. The sixteen-core Zen 5c CCD also has 32 MB L3, yielding 384 MB across twelve CCDs.

Unlike Bergamo's two-complex CCD, the EPYC 9005 architecture overview describes one complex per CCD for both Turin variants. This changes the local sharing domain and is an important detail that core-count summaries omit.

Hot Chips 2024 describes Zen 5's expanded front end, execution resources and memory subsystem. The 48 KB L1 data cache, larger execution capacity and increased data bandwidth support more work per core. In the server implementation, full-width 512-bit floating-point execution strengthens vector capability. The ISSCC 2025 paper complements this with the 4 nm core-complex implementation and explains why configurations across product segments need to be distinguished.

The EPYC-specific architecture guide adds a less prominent but valuable detail: selected “wide” products connect each CCD to two GMI links rather than one, increasing the bandwidth between the compute chiplet and I/O die. Package DRAM bandwidth alone therefore does not fully describe the resources available to a particular core population.

Turin launched at Advancing AI in October 2024. Google Cloud Next 2025 connected the generation to C4D and H4D virtual machines. Those cloud disclosures add workload and deployment context; the core and package organization still come from Hot Chips, ISSCC and the EPYC technical documents.

At maximum dense-core count, memory remains a key balance: 614.4 GB/s of nominal DDR5 bandwidth divided among 192 cores is 3.2 GB/s per core. The same memory interface divided among 128 cores is 4.8 GB/s per core. The dense version can deliver more aggregate work when the workload supports that tradeoff, but more cores do not guarantee proportionally more memory throughput.

1.8 Venice: EPYC 9006, 2026

By the report cutoff, Venice had moved beyond roadmap previews. AMD launched the sixth-generation EPYC portfolio at Advancing AI on 23 July 2026 and published EPYC 9006 product specifications. The family uses Zen 6 and Zen 6c and expands into SP7 and SP8 platforms. Production ramp and partner validation are documented, but a portfolio-wide GA date is not; this report classifies the new generation as limited availability at the cutoff.

The flagship EPYC 9996 specification provides a concrete reference point: 256 cores, 512 threads, 1,024 MB L3, sixteen memory channels and PCIe 6.0 x96 on SP7. AMD lists DDR5 RDIMM rates up to 8,000 MT/s and MRDIMM rates up to 12,800 MT/s, corresponding to approximately 1.024 TB/s and 1.638 TB/s per socket. Its stated default CPU power is 600 W with a 400–600 W range.

The platform change matters more than the core-count increment alone. Relative to SP5, the flagship adds memory channels and a new I/O generation. AMD's Taiwan ecosystem announcement also identifies 2.5D Elevated Fanout Bridge packaging for Venice, extending advanced-package methods into this CPU generation.

AMD presents SP8 for enterprise and general-purpose systems, alongside SP7's larger platform. Its portfolio page also identifies 9006X and an AI-host-oriented 9006 LP branch. Detailed availability and specifications should be read by individual product rather than assumed to be identical across the entire family.

Venice also serves a system role in Helios, whose announced configuration combines eighteen EPYC CPUs with seventy-two Instinct MI455X GPUs. The CPU is responsible for host-side execution and data movement in a much larger accelerator system.

The 2026 public sources reviewed here provide a firmer basis for describing the platform than for reconstructing every Zen 6 internal structure. The Hot Chips 2026 program contains AMD's MI400 architecture and system talks, but does not establish an equivalent Zen 6 CPU disclosure. This report therefore does not assign unverified pipeline or cache-topology details to Venice.

1.9 Smaller servers, edge systems and a custom HBM branch

EPYC 4004 and 4005 extend the server brand into AM5 systems with up to sixteen cores. The former uses Zen 4 and the latter Zen 5. Their close relationship to the desktop platform is useful for understanding their economics and core technology, while server validation and the intended deployment distinguish the products. They do not offer the memory-channel and I/O scale of SP5 EPYC.

EPYC 8005, introduced in 2026 and codenamed Sorano in AMD's architecture guide, advances the SP6 branch to up to 84 Zen 5 cores. AMD specifies six DDR5-6400 channels, 96 PCIe 5.0 lanes and a 70-225 W product range. The 84-core 8635P has 384 MB L3, arranged as twelve CCDs with seven enabled cores and 32 MB L3 per CCD; the guide specifies 1 MB L2 per core and up to 48 lanes of CXL 2.0. Its deployment focus includes telecommunications, edge services and storage. This is a continuation of platform specialization, rather than a replacement for the larger EPYC line.

A particularly instructive branch is the custom EPYC implementation behind Microsoft's Azure HBv5. The SCA/HPCAsia 2026 paper describes CPU and cloud co-design for memory-bandwidth-bound HPC, using HBM with a Genoa-based design. Microsoft's current VM documentation lists 368 usable CPU cores, 432 GB of memory and 6.7 TB/s of memory bandwidth for the largest HBv5 configuration. These are VM-level resources, not specifications for a retail single-socket EPYC.

HBv5 demonstrates that the technology shared by EPYC and Instinct extends beyond CPU core IP: advanced packaging and HBM can also support a CPU-centric cloud system. Its architecture belongs in the EPYC story, but should not be conflated with MI300A or a generally available socketed Genoa SKU.

1.10 What changed in memory bandwidth per core?

The following arithmetic isolates one resource balance. It uses nominal maximum memory rates, fully populated channels and the representative maximum core counts shown. Bandwidth is theoretical payload bandwidth before protocol, access-pattern and software effects; these are not benchmark results.

Configuration	Channels and data rate	Calculated GB/s per socket	Cores	Calculated GB/s per core
Naples	8 × DDR4-2667	170.7	32	5.33
Rome / Milan	8 × DDR4-3200	204.8	64	3.20
Genoa	12 × DDR5-4800	460.8	96	4.80
Bergamo	12 × DDR5-4800	460.8	128	3.60
Siena	6 × DDR5-4800	230.4	64	3.60
Turin, Zen 5	12 × DDR5-6400, current	614.4	128	4.80
Turin, Zen 5c	12 × DDR5-6400, current	614.4	192	3.20
Sorano	6 × DDR5-6400	307.2	84	3.66
Venice 9996, RDIMM	16 × DDR5-8000	1,024.0	256	4.00
Venice 9996, MRDIMM	16 × DDR5-12800	1,638.4	256	6.40

Calculation: channels × transfers per second × 8 payload bytes per channel. The table shows why cache-rich products, higher-bandwidth memory and fewer-core high-frequency products remain useful

even as maximum core count rises. It also shows why a simple ranking by cores misses much of the engineering story.

1.11 Cache-sharing domains across EPYC generations

Product	Full compute-die organization	L3 within one sharing domain	Maximum package L3
Naples	2 complexes × 4 cores per Zeppelin SoC	8 MB / 4 cores	64 MB
Rome	2 complexes × 4 cores per CCD	16 MB / 4 cores	256 MB
Milan	1 complex × 8 cores per CCD	32 MB / 8 cores	256 MB
Milan-X	1 complex × 8 cores, with stacked cache	96 MB / 8 cores	768 MB
Genoa	1 complex × 8 cores per CCD	32 MB / 8 cores	384 MB
Genoa-X	1 complex × 8 cores, with stacked cache	96 MB / 8 cores	1,152 MB
Bergamo	2 complexes × 8 cores per CCD	16 MB / 8 cores	256 MB
Turin, Zen 5	1 complex × 8 cores per CCD	32 MB / 8 cores	512 MB
Turin, Zen 5c	1 complex × 16 cores per CCD	32 MB / 16 cores	384 MB

The table describes full architectural complexes; individual SKUs may disable cores or populate fewer chiplets. Package cache totals are sums of distributed L3 resources. The progression is not monotonic in cache per core: the cloud-density branches intentionally make different choices. Milan and Turin Zen 5c also show why the number of cores sharing one L3 must be tracked independently from the total cores on a die.

1.12 Core implementation: feeding wider execution

The change from Zen to Zen 5 is more than an ISA progression. Zen established SMT, an op cache, out-of-order execution and a private-cache hierarchy that could be replicated in a CCX. Zen 2 widened floating-point execution to 256 bits. Zen 3 enlarged the execution window and load/store capacity while joining eight cores behind one L3. Zen 4 added a larger private L2 and AVX-512 instructions without making the entire floating-point path 512 bits wide.

Zen 5 makes the front-end and data-path expansion explicit. Its two fetch/decode paths feed eight-wide dispatch, while a larger reorder window lets the core find more independent work. The Hot Chips slides disclose a 448-entry retire/ROB capacity in single-thread mode and 224 entries per thread in two-thread mode. Six ALUs and four AGUs provide more execution opportunity, but sustained utilization still requires predictable branches, independent operations and data delivery.

The server implementation's 48 KB L1 data cache retains a four-cycle load-to-use target. The core can handle four memory addresses per cycle, with a maximum of two 512-bit loads or one 512-bit store through the vector-width paths. The published two-load/one-store capacities should not be read as all independent maxima occurring simultaneously without resource constraints. Wider L2-to-core paths and

a new two-dimensional stride prefetcher address the feeding problem. L3 retains a victim relationship to L2, with duplicated L2 tags used for probe filtering.

Core	Front end and dispatch	Execution window	Load/store and vector	Local caches	Sources
Zen	32 B fetch; 4 x86 decode; 6 micro-op dispatch	192-entry retire queue; SMT2; 72-load / 44-store queues	2 loads + 1 store resources; 128-bit FP	64 KB I; 32 KB D; 512 KB L2; 8 MB L3/4 cores	A1 A3
Zen 2	Larger op cache and improved prediction; 6-wide dispatch	224-entry ROB in Zen 3 comparative disclosure	256-bit FP; baseline 2 loads/1 store per cycle	32 KB I; 32 KB D; 512 KB L2; 16 MB L3/4 cores	A5 A6 B2
Zen 3	4-decode / up to 8 op-cache ops; 6 macro-op dispatch	256-entry ROB; larger integer/FP issue capacity	3 loads/2 stores; 256-bit FP; FMA latency 4 cycles	32 KB I; 32 KB D; 512 KB L2; 32 MB L3/8 cores	B1 B2 B3
Zen 4 / 4c	Larger op cache; improved branch prediction and dispatch	Retire and load/store queues enlarged; numeric sizes n/d in reviewed slides	AVX-512 on 256-bit resources; BF16/VNNI	32 KB I; 32 KB D; 1 MB L2; L3 32/16 MB per 8-core CCX	B8 B9 B10
Zen 5 server	2 x32 B fetch; 2 x4 decode; 6K op cache; 8-wide dispatch	ROB 448 entries 1T / 224 per thread 2T; 240 integer physical registers	6 ALU, 4 AGU; 4 memory addresses/cycle; 2 x512-bit loads or 1 x512-bit store capacity	32 KB I; 48 KB 12-way D, 4-cycle load-to-use; 1 MB 16-way L2	C1 C2 C3
Zen 6 / 6c	n/d in reviewed public technical record	n/d	n/d; avoid deriving ISA from branding	9996 package L3 1024 MB; private caches and sharing topology n/d	C8 N30 N31

Zen 4c and Zen 5c share an ISA foundation with their larger counterparts, but physical design and cache sharing change the system balance. Hot Chips reports about 35% less core-plus-L2 area for Zen 4c than Zen 4. That is a disclosed implementation comparison, not a universal area or performance multiplier. Dense products make sense when aggregate throughput and energy at the intended operating point outweigh lower per-core cache capacity or frequency.

1.13 Fabric, memory configuration and the accelerator-host role

The I/O die makes the socket easier to scale, but it is also a shared traffic junction. A core miss crosses a CCD link before reaching a memory controller; a remote-socket access adds xGMI. NPS settings describe how memory-controller locality is exposed, while the CCX describes a cache-sharing domain. These are different partitions. Increasing the number of NUMA nodes can help locality-sensitive software and complicate placement at the same time.

SP5 systems can allocate high-speed links between PCIe and socket connectivity. The common single-socket maximum is 128 PCIe lanes; a two-socket configuration does not simply double that number because socket links consume resources. The Turin guide gives a 160-lane system configuration with three xGMI links. CXL-capable lanes are a subset of the I/O budget, not extra lanes added to the headline total. A CXL memory device also has a different latency and bandwidth path from local DDR.

An AI host therefore needs to be selected using the actual board topology: which CPU owns each GPU or NIC, how PCIe switches are arranged, whether peer DMA is supported on that route, and how host memory is populated. A processor with more cores can be a worse fit for a GPU host if it leaves insufficient I/O or memory bandwidth for the required traffic. Conversely, CPU inference and

preprocessing may benefit from AVX-512, BF16/VNNI, cache capacity and frequency even when no discrete GPU is present.

Security and RAS evolve alongside performance. Naples introduced the EPYC SEV lineage; later implementations added protection of guest execution state, and Milan introduced SEV-SNP's stronger integrity model. The current Infinity Guard and SEV documentation must be read with platform firmware and hypervisor requirements. Memory ECC, error containment and server validation belong to the usable system, not merely the core microarchitecture.

CPU power ratings do not specify one sustained frequency for every workload. Core population, instruction mix, memory traffic, thermal conditions and configured power limits determine the operating point. High-frequency, cache-rich, dense-core and low-power SKUs deliberately sit at different points on that curve. The family tables report maxima; selecting a concrete system requires its exact OPN and memory configuration.

1.14 Implementation limits in the public record

The Sorano architecture guide contains conflicting 4 nm and 5 nm statements and inconsistent memory-capacity wording relative to the data sheet. Its topology and cache capacity are independently corroborated by the product specifications; its process node is left unresolved here. The guide's single-socket product context also takes precedence over generic socket-link language copied into the document.

For Venice, the reviewed record establishes the N2 compute direction, EFB packaging and published socket resources. It does not yet support a full Zen 6 pipeline, private-cache, coherence-directory or die-to-die energy reconstruction. This is a specific disclosure gap, not a reason to infer those structures from Zen 5 or from the 1,024 MB package-cache total.

Source map: A1-A10; B1-B15; C1-C12; G1, G4, G13, G18; N1-N4, N20, N22, N23, N30-N33, N39, N45-N48, N50. Full references appear in the bibliography; numerical cells are indexed in the companion claims ledger.

2. Instinct products and technology evolution

2.1 The product sequence

Product	Introduction	Architecture	Representative memory	Principal change
Radeon Instinct MI6 / MI8 / MI25	Announced 2016; 2017 generation	Polaris / Fiji / Vega	16 GB GDDR5 / 4 GB HBM / 16 GB HBM2	Compute portfolio assembled from graphics-derived architectures
Radeon Instinct MI50 / MI60	2018	Vega 20	MI60: 32 GB HBM2, about 1 TB/s	7 nm, stronger FP64 and GPU-to-GPU fabric
Instinct MI100	2020	CDNA	32 GB HBM2, 1.23 TB/s	Dedicated compute architecture and matrix execution
Instinct MI250 / MI250X	2021	CDNA 2	128 GB HBM2e, 3.2 TB/s per package	Two GPU compute dies and exascale node integration
Instinct MI210	2022	CDNA 2	64 GB HBM2e, 1.6 TB/s	Single-die PCIe branch
Instinct MI300A	2023	Zen 4 + CDNA 3	128 GB shared HBM3, 5.3 TB/s	CPU and GPU chiplets share coherent physical memory
Instinct MI300X	2023	CDNA 3	192 GB HBM3, 5.3 TB/s	Eight stacked GPU compute chiplets, optimized for accelerator systems
Instinct MI325X	2024	CDNA 3	256 GB HBM3e, 6.0 TB/s	Memory-capacity and bandwidth refresh
Instinct MI350X / MI355X	2025	CDNA 4	288 GB HBM3e, 8.0 TB/s	Stronger matrix engines, low-precision formats and revised I/O partitioning
Instinct MI350P	2026	CDNA 4	144 GB HBM3e, 4.0 TB/s	PCIe card for conventional server infrastructure
Instinct MI455X, MI400 family	2026	CDNA 5	432 GB HBM4, 23.3 TB/s	New execution organization and rack-scale Helios integration

Memory bandwidth is per accelerator package or card. MI250X contains two software-visible GPU devices, so its package totals should not be confused with the resources visible to one device. Published memory capacities follow AMD's product naming convention.

2.2 The prelude and first Radeon Instinct generation, 2015-2017

The first Radeon Instinct portfolio was announced in December 2016. MI6, MI8 and MI25 drew on Polaris, Fiji and Vega respectively. The product name united devices with materially different memory systems: GDDR5 for MI6, first-generation HBM for MI8 and HBM2 for MI25.

Fiji's earlier use of HBM is important background. Placing wide memory interfaces beside a GPU in an advanced package established a practical route to high bandwidth without relying solely on ever-faster external DRAM interfaces. This packaging and memory lineage continued into Instinct even as the compute architecture changed.

MI25's Vega architecture remains closely connected to Radeon Vega graphics. Hot Chips 2017's Vega presentation explains the programmable compute organization, packed arithmetic and memory-system ideas behind that generation. The server product additionally depended on accelerator packaging, cooling, reliability and software.

At this stage, the story is one of adapting a broad GPU architecture to compute markets. It should not be retroactively described as CDNA. AMD's ROCm software effort was also becoming an essential part of the platform: an accelerator's usefulness depended on compiler, runtime and library support as well as peak arithmetic rates.

2.3 MI50 and MI60: Vega 20, 2018

MI50 and MI60 arrived at AMD's Next Horizon event in November 2018. Vega 20 moved the compute product to 7 nm and strengthened the HPC proposition through high-rate FP64 arithmetic, HBM2 bandwidth around 1 TB/s, PCIe 4.0 and direct GPU-to-GPU Infinity Fabric connectivity. MI60 provided 32 GB of HBM2; MI50 existed in more than one memory-capacity configuration.

The important change was broader than process scaling. Scientific workloads needed double-precision arithmetic and reliable movement of large data sets, while multi-accelerator applications needed communication paths that avoided unnecessary host-mediated transfers. MI50/MI60 addressed those requirements while retaining Vega ancestry.

Radeon VII used related Vega 20 technology in a desktop product. That connection is useful for tracing silicon reuse, but the compute products' validated capabilities and server configuration should be taken from Instinct documentation.

Next Horizon is a particularly informative industry event in this history: it combined the 7 nm GPU announcement with Rome's server direction. Both product lines were beginning to turn leading-edge process technology and fabric connectivity into data-center platforms.

2.4 MI100: the first CDNA generation, 2020

MI100 established CDNA as AMD's dedicated data-center compute architecture. It combined 120 compute units with 32 GB of HBM2 and approximately 1.23 TB/s memory bandwidth. Its matrix instructions and execution resources made matrix multiplication a first-class architectural function rather than relying solely on conventional vector instructions.

The separation from RDNA is central. RDNA developed around graphics and gaming priorities; CDNA developed around compute, matrix arithmetic, scientific precision and scaling. Both are GPU architectures, but conclusions about execution width, caches or instruction behavior cannot be transferred from one to the other simply because both carry AMD branding.

The CDNA white paper explains the compute-unit and matrix organization. AMD's November 2020 announcements, aligned with the HPC community's SC season, place MI100 in an EPYC-based system and software context. The generation retained a large single GPU compute die, so it is the last major Instinct step before compute itself was split across dies.

For historical comparisons, MI100 is a useful starting point for the dedicated compute lineage. It is less useful to compare its peak FP64 number directly with a later generation's FP8 or FP4 headline: those measure different arithmetic and often different execution paths.

2.5 MI200: CDNA 2, MI250/MI250X and MI210, 2021-2022

MI250 and MI250X moved Instinct to a two-die GPU package. MI250X combined two 6 nm graphics compute dies, or GCDs, with eight HBM2e stacks, 128 GB total memory and 3.2 TB/s aggregate memory bandwidth. The package had 220 active compute units. MI210 brought CDNA 2 to a single-die PCIe card with 64 GB and 1.6 TB/s.

A crucial software detail is that MI250X's two GCDs are exposed as two GPU devices. One physical accelerator module is therefore not automatically one flat GPU programming domain. Work assignment and communication between the two devices remain part of application design.

CDNA 2 strengthened matrix computation, including high-precision work relevant to HPC, and developed coherent CPU/GPU connectivity. Coherence and unified addressing, however, do not make host DDR memory and GPU HBM one physical memory pool. Their location, bandwidth and capacity remain distinct.

Hot Chips 2022 is the key architectural disclosure because it discusses both the MI200 accelerator and node organizations. Frontier's node combines one custom EPYC CPU with four MI250X modules, yielding eight GPU devices. The topology, including accelerator and network connections, is part of how the machine sustains useful scaling.

The VLSI 2023 paper on MI250X's Elevated Fanout Bridge packaging supplies an implementation complement. It explains the packaging technology that allows the two large compute dies and their HBM resources to form a practical accelerator. ISCA 2023's exascale retrospective then explains the path from research alternatives to the delivered Frontier architecture.

Reading these together avoids an incomplete interpretation of Frontier as merely a collection of fast GPUs. CPU/GPU interaction, memory access, node networking, power and application co-design all shaped the final system. Frontier also provided a major software and operational proving ground before the more integrated MI300A design.

2.6 MI300A: a CPU-GPU processor with shared HBM, 2023

MI300A combines twenty-four Zen 4 CPU cores with CDNA 3 GPU compute and 128 GB of HBM3 in one package. Its compute layer contains three CPU CCDs and six accelerator complex dies, or XCDs, over four I/O dies. The CPU and GPU share a coherent physical HBM memory system with approximately 5.3 TB/s aggregate bandwidth.

That physical integration changes the programming problem. A CPU phase and GPU phase can operate on shared data without requiring the same explicit host-DRAM-to-device-HBM copy pattern as a conventional discrete accelerator. Synchronization and efficient data access still matter; shared memory is a capability, not an automatic performance guarantee.

The design draws directly on both EPYC and Instinct. The Zen 4 CCDs adapt the CPU chiplet for stacked integration, while the GPU XCDs are designed for the vertical interface. ISCA 2024 explains how AMD accommodated the different chiplet footprints and interfaces on a reusable I/O-die design. It also explains the engineering history: earlier exascale APU concepts were constrained by packaging, manufacturing and thermal practicality, while MI300A could draw on more mature integration techniques.

ISSCC 2024's MI300 package paper supplies the implementation view. The package combines leading-edge compute, infrastructure dies and HBM rather than building everything as one enormous logic die. The same general assembly strategy can support different mixtures of CPU and GPU compute.

El Capitan turns that architectural idea into a system. LLNL documents four MI300A APUs per compute node, giving ninety-six CPU cores and 512 GB of shared-package memory across the node. SC24 marked the system's major public performance milestone. This deployment provides context for why MI300A was designed as a heterogeneous HPC processor, even while the wider MI300 family also pursued AI.

The deepest academic reading sequence for this product is HPCA 2017's exascale APU proposal, ISCA 2023's Frontier retrospective and ISCA 2024's account of realizing the heterogeneous processor. The sequence explains both the continuity of the idea and the changes required to make a commercial implementation possible.

2.7 MI300X: CDNA 3 for large accelerator systems, 2023

MI300X uses the MI300 integration platform for a different balance: eight GPU XCDs, four I/O dies and eight HBM3 stacks provide 304 active compute units, 192 GB of memory and approximately 5.3 TB/s bandwidth. The CPU remains external to the accelerator.

This is the key distinction from MI300A. Both products use closely related packaging and compute technology, but only MI300A integrates Zen CPU cores into the shared-HBM processor. MI300X's large HBM does not, by itself, turn the host CPU's memory into a single physical pool with the accelerator.

Hot Chips 2024 explains MI300X at the package and platform levels. In its eight-accelerator baseboard, each GPU has direct connections to the other seven. The presentation's aggregate 896 GB/s per-GPU interconnect figure is bidirectional. It must not be compared without adjustment to a one-direction bandwidth number from another system.

MI300 also changes the relationship between physical chiplets and logical devices. The GPU can be presented as one large device across multiple XCDs, while supported partitioning modes divide resources for other deployment needs. Compute partitioning and memory partitioning are related but distinct configuration choices.

The combination of ISSCC 2024, ISCA 2024 and Hot Chips 2024 provides an unusually complete picture: physical construction, the rationale for the heterogeneous platform, and the organization of the AI-oriented accelerator and baseboard. The later IEEE Micro article on MI300X interconnect design adds detail; it is a journal article, not a paper at the MICRO conference.

MI300X launched at Advancing AI in December 2023. Microsoft Ignite 2023 had announced Azure's preview direction; Microsoft Build 2024 subsequently supplied a deployment and availability milestone for Azure's MI300X virtual machines and services.

2.8 MI325X: more memory within CDNA 3, 2024

MI325X retained CDNA 3 while moving to 256 GB of HBM3e and 6 TB/s memory bandwidth. It is a memory and platform refresh rather than a new CDNA generation.

The capacity increase matters to AI serving and training because model weights, runtime state, activations and attention caches compete for HBM. A larger per-GPU memory budget can change how a model is partitioned or how much concurrency fits on a device. The effect depends on the model and execution software, rather than following directly from capacity alone.

AMD previewed MI325X at Computex 2024 and launched it at Advancing AI in October. The final product specification is the appropriate comparison point: the early roadmap figure of 288 GB was superseded by the launched 256 GB configuration.

2.9 MI350X and MI355X: CDNA 4, 2025

CDNA 4 substantially reworked the compute and infrastructure balance. MI350-series accelerators use eight 3 nm XCDs above two 6 nm I/O dies, compared with MI300's four I/O dies. Keeping infrastructure on a different process remained part of the strategy, but the infrastructure partition itself changed.

The compute units became more capable while the active count fell to 256 across the package. This is a useful warning against treating CU count as a stable measure of work across architectures. Matrix resources, instruction support, clocking and feeding the execution machinery matter more than the count alone.

CDNA 4 adds low-precision microscaling formats, including MXFP4 and MXFP6, alongside established AI and scientific formats. These formats trade numerical representation against throughput and data movement. They can benefit appropriate AI workloads, but an FP4 rate is not a replacement for an FP64 scientific-computing metric.

Both MI350X and MI355X provide 288 GB HBM3e and 8 TB/s memory bandwidth. AMD positions MI350X as a 1,000 W air-cooled option in an eight-GPU baseboard ecosystem, while MI355X uses direct liquid cooling at a 1,400 W rating. Power delivery and cooling are therefore explicit product choices.

Advancing AI 2025 introduced the products and software direction. Hot Chips 2025's CDNA 4 presentation explains the architecture and platforms. ISSCC 2026 follows with the implementation paper on stacked 3 nm XCDs and 6 nm I/O dies. The year difference is normal: a product launch, architecture presentation and circuit paper need not occur in the same calendar year.

MI350P extends CDNA 4 into a PCIe card in 2026, with 144 GB HBM3e and 4 TB/s. It uses four XCDs, one I/O die and four HBM stacks, with 128 active CUs. It addresses conventional PCIe servers with up to 600 W board power and a configurable 450 W point. AMD published a detailed deployment discussion on 7 May 2026, before featuring the product again at Advancing AI in July. A dedicated peer-fabric configuration is not specified for the card in the reviewed brief, so an eight-card PCIe server should not be assumed to have the UBB's all-to-all Infinity Fabric.

2.10 MI400 and MI455X: CDNA 5 and Helios, 2026

AMD's July 2026 Advancing AI event launched MI400 and Helios and described production activity. Partner validation and scheduled customer deployments show that launch, production and broad availability are distinct milestones; the new platform is classified as limited availability here. The public specifications identify MI455X as the first CDNA 5 product, with 432 GB HBM4 across twelve stacks and up to 23.3 TB/s memory bandwidth.

The architectural change goes beyond replacing HBM3e with HBM4. AMD describes eight XCDs with 256 work-group processors, or WGPs, and Wave32 execution. It also separates fabric/cache dies from I/O dies: two fabric/cache dies provide the HBM interface and 192 MB global L2, while two I/O dies handle external connectivity. This differs from the MI300 and MI350 infrastructure partitioning.

The published CDNA 5 description also introduces mechanisms aimed at moving data and coordinating work more efficiently, including asynchronous transfers into local data share and cooperation across WGPs. These changes make the programming and scheduling organization as important as the arithmetic peak.

Hot Chips 2026 includes separate talks on MI400's GPU architecture and its system architecture. That pairing reflects the product's scope. Helios combines seventy-two MI455X GPUs and eighteen Venice CPUs, with scale-up communication using UALink over Ethernet and separate scale-out networking. AMD specifies approximately 31 TB of aggregate HBM across the rack. That aggregate capacity remains distributed across accelerators and should not be read as one uniformly accessible local memory.

The rack product also brings power delivery, liquid cooling, serviceability, networking and software into the architectural discussion. OCP 2025 exposed the open-rack design direction before the 2026 launch, connecting AMD's GPU roadmap to Meta's rack-design ecosystem.

MI430X is an announced HPC/sovereign-AI branch with release planned for 2027, and MI500 is a future roadmap generation. Their existence reinforces the need to distinguish family branding from identical enabled features and availability. They appear as announced or roadmap rows in the comparisons, with undisclosed fields left empty of inferred specifications.

2.11 Matrix engines, local storage and numerical formats

MI100's matrix engine is the first useful baseline for the dedicated compute line, but its BF16 and FP16 rates differ: the published BF16 peak is 92.3 TFLOP/s, while FP16 reaches 184.6 TFLOP/s. Treating these as interchangeable would distort the transition to MI250X's 383 TFLOP/s BF16. CDNA 2 also adds matrix FP64, making the distinction between vector and matrix double precision essential.

CDNA 3 adds FP8 formats and stronger matrix throughput, while retaining a Wave64 execution model. Its 4 MB L2 per XCD and package-level Infinity Cache sit between CU-local storage and HBM. The partition is deliberately hierarchical: a large package cache total does not imply that all operand traffic sees the same path or latency.

CDNA 4 enlarges LDS from 64 KB to 160 KB per CU and supports direct loading into LDS, reducing the need to stage data through vector registers. This helps kernels retain tiles and overlap movement with computation. It adds OCP microscaling formats, including MXFP8, MXFP6 and MXFP4, whose block scales and representation affect numerical behavior as well as storage. Native TF32 hardware support is replaced by a software path based on BF16 operations; software compatibility therefore does not imply identical execution hardware.

Architecture	Execution and matrix change	LDS per CU	Vector L1	L2 / memory-side cache	Sources
CDNA / MI100	Wave64; MFMA matrix operations; BF16 peak differs from FP16	64 KB	16 KB	8 MB L2 per GPU	D5 N27 E19
CDNA 2 / MI200	Wave64; adds FP64 matrix execution; MI250X has 2 GPU devices	64 KB	16 KB	8 MB L2 per GCD; 16 MB per MI250X package	D7 D8 N26 E19
CDNA 3 / MI300	Wave64; FP8 E4M3/E5M2; one logical device can span XCDs	64 KB	32 KB	4 MB L2/XCD; 256 MB Infinity Cache/package	E1 E3 N25 E19
CDNA 4 / MI350	Wave64; MXFP8/6/4; direct global-to-LDS path; native TF32 replaced by BF16-based software path	160 KB	32 KB	4 MB L2/XCD; 256 MB Infinity Cache for X, 128 MB for P	E10 N24 E19
CDNA 5 / MI455X	Wave32 WGPs; Tensor Data Mover, L2 multicast and WGP cooperation	n/d here; WGP scope differs	n/d here	192 MB global L2 on two fabric/cache dies	E15 K3 N14

MI350X and MI355X deliberately use different frequency/power points. The latter's dense BF16 peak rises to 2,517 TFLOP/s from 2,307, while rated TBP rises from 1,000 W to 1,400 W. The higher-power device does not increase HBM capacity or bandwidth. This is a throughput-density choice that must be assessed with rack limits and measured workload efficiency, not only peak arithmetic.

CDNA 5 changes the execution unit to a Wave32 WGP organization. Tensor Data Mover transfers can feed LDS asynchronously, L2 multicast can reuse a transfer across cooperating work, and named barriers

support finer coordination. These mechanisms address data movement and synchronization around matrix computation. A count of 256 WGP's cannot be equated to the 256 CUs of CDNA 4.

The precision trade becomes unusually explicit in MI455X. Its dense BF16 peak is 5,033 TFLOP/s, OCP FP8/MXFP8 reaches 20,133, MXFP6 is 20,133, and MXFP4 is 40,265. Published vector and matrix FP64 are each 5 TFLOP/s. This is an AI-oriented balance, with the announced MI430X HPC branch targeting up to 288 TFLOP/s hardware FP64 in 2027. No unpublished MI430X memory or packaging specifications are inferred from MI455X.

2.12 Physical partitions, logical devices and isolation

MI250X's two logical GPUs each own their local HBM and L2. The package totals of 128 GB and about 3.2 TB/s require work and traffic to be considered across both GCDs. The bridge and internal fabric create a fast connection; they do not erase locality. In Frontier, node-level CPU and network connections further constrain how traffic can be scheduled.

MI300X and MI350 can expose a larger logical GPU across multiple compute chiplets or divide resources using supported compute and memory partition modes. MI350's two I/O dies support NPS1 and NPS2 memory organizations. In the documented NPS2/DPX arrangement, a logical partition groups four XCDs with 144 GB of HBM. SPX, DPX, QPX and CPX describe compute partition choices; they are not interchangeable names for memory partitioning.

A resource partition is also not automatically a security guarantee. Runtime support, virtualization mode, address translation, firmware and the particular product determine isolation behavior. Capacity planning should use the usable partition and supported software configuration rather than the whole-package number. The companion software chronology separates hardware introduction from release-specific enablement.

The packaging progression serves different needs at each step. MI250X uses EFB to join two large compute dies. MI300 stacks compute over four active I/O dies, enabling both a CPU–GPU APU and a GPU-only product. MI350 reduces that infrastructure to two I/O dies while increasing matrix resources and HBM performance. MI455X separates fabric/cache and external-I/O functions. These are changes in the partition of the machine, not simply successive shrinks of one block diagram.

2.13 Regional products and the forward roadmap

AMD's Q2 2026 filing identifies MI308 as a license-dependent regional product and reports resumed licensed shipments at the end of fiscal 2025. It also discusses MI325 licenses for some customers. The reviewed filing does not provide a technical MI308 specification table, so this report records the branch and its availability constraint without inventing compute, memory or interconnect limits. Export-control conditions explain why the branch exists; they do not establish a particular architectural cap absent a primary specification.

The July 2026 roadmap names MI500/Helios 500 for 2027 and MI600/Helios 600 for 2028. The corresponding host and networking codenames are recorded in the crosswalk. Those entries carry their disclosure date and remain roadmap items. Full-stack diagrams for 2026 use the published MI455X configuration; they do not silently incorporate features promised for the following generations.

Remaining material gaps include the final MI455X board-power envelope, detailed per-die node assignment, some HBM stack heights, and per-physical-link UALoE bandwidth. Package-level rates are published, but dividing them into a presumed lane topology would create unsupported precision. ISSCC 2026 provides a MI350 implementation record; a comparable MI400 ISSCC implementation paper is not established in this review.

Source map: D1–D10; E1–E20; F2, F8, F9; K3; N5–N16, N24–N29, N35–N37, N40, N41. Full references appear in the bibliography; numerical cells are indexed in the companion claims ledger.

3. Scale-up interconnect and memory semantics

3.1 One brand, several physical and logical interfaces

Infinity Fabric names a family of connections, not one fixed wire speed or coherence contract. CPU CCD-to-IOD links, CPU socket-to-socket xGMI, GPU die-to-die links, GPU peer links and coherent CPU-GPU connections must be compared at their own endpoints. A rate measured across one of these boundaries is not a substitute for another.

Naples used the fabric to join complete SoCs inside a socket and to connect two sockets. Rome changed the on-package topology by introducing a central I/O die. Genoa and Turin increase external signaling and retain configurable socket/I/O allocations. Their NUMA behavior follows where memory controllers and cache domains sit, even when software sees one socket.

In Instinct, Vega 20 introduced direct peer connectivity and MI100 supported a four-GPU hive. MI200 added package-interior connectivity between two GCDs and node-level configurations with CPU and GPU links. MI300X and MI350 use a direct eight-GPU all-to-all baseboard: each GPU has seven peer connections. This provides one-hop peer paths within the baseboard but does not create an arbitrary-size domain by adding more identical boards.

3.2 Normalize the link before reasoning about the system

MI300X's seven peer links each provide 64 GB/s in one direction, giving 448 GB/s per GPU per direction. The familiar 896 GB/s number adds transmit and receive. MI350 increases each peer link to 76.8 GB/s, giving 537.6 GB/s in one direction or 1,075.2 GB/s bidirectionally. Both remain eight-GPU baseboard domains.

MI250X requires a different accounting boundary. Its internal GCD connection is quoted at 400 GB/s bidirectionally, or 200 GB/s in each direction between the two compute dies. Its external IF links are separate connections, each 50 GB/s per direction. Summing those ports describes interface capacity, not all-to-all bisection or bandwidth available to one collective. CPU attachment and node routing also consume or constrain connections.

For CPU xGMI, multiplying lane count by signaling rate yields an electrical ceiling. A sixteen-lane link at 32 Gb/s per lane gives 64 GB/s per direction before protocol effects. The table labels this as raw arithmetic rather than a sustained cache-coherent payload result. Likewise, a PCIe 5 x16 encoding ceiling near 63 GB/s is not a measured DMA throughput guarantee.

3.3 Helios: a switched seventy-two-GPU domain

Helios moves scale-up outside the eight-GPU direct mesh. AMD describes UALink over Ethernet, or UALoE, using a single-hop, multiplane fabric with twelve switches in six trays. The seventy-two GPUs are distributed across eighteen four-GPU compute trays with eighteen Venice CPUs. The published 3.6 TB/s per-GPU scale-up number is bidirectional: the normalized value is 1.8 TB/s per direction.

The switch makes the domain larger without requiring one dedicated direct cable to every peer. Its cost is a switching, buffering, congestion and operational layer. Reach, physical cabling and failure domains become rack-design questions. The reviewed final blueprint does not establish every switch radix, per-lane electrical rate or reduction-offload feature, so this report does not fill those fields from a generic UALink standard or another vendor's switch.

GPU memory semantics do not imply that all rack memory has local-HBM latency or full CPU cache coherence. Remote accesses still traverse endpoint and network resources. RCCL and application

partitioning must organize collective and point-to-point traffic around those constraints. Fabric Manager, Fabric OS and vPods add provisioning and operational boundaries above the physical fabric.

3.4 Open standards and the remaining integration work

UALink addresses scale-up accelerator communication; UEC addresses Ethernet-based scale-out; CXL addresses coherent host/device and memory attachment; OCP addresses system interfaces and designs. They operate at different layers. AMD's participation and adoption do not make their names interchangeable, nor does an “UEC-ready” NIC statement prove certification of every protocol feature.

Helios uses partner switching and physical-network components. No standalone AMD merchant datacenter switch ASIC, optical-module family or CPO product is established in the reviewed portfolio. Switches and optics are therefore explicit ecosystem components in the stack figures. This still leaves AMD responsible for a much larger integration surface than the GPU package alone.

Source map: A2, A4, A9, B11, C3, D3, D5, D7, D8, E1, E3, E10, E15–E18, K2–K4; N14, N15, N24–N27. Full references appear in the bibliography; numerical cells are indexed in the companion claims ledger.

4. Networking: DPUs, AI NICs and related adapters

4.1 The product sequence and its place in AMD

AMD completed its acquisition of Pensando on 26 May 2022. Capri and Elba therefore began as Pensando designs before becoming part of AMD's portfolio. Their history belongs in this report because it establishes the architecture that AMD subsequently extended through Giglio and Salina.

A Pensando **data processing unit** executes infrastructure services: virtual networking, connection tracking, security, storage protocols and telemetry. A distributed services card, or DSC, is a board containing the DPU and its supporting components. The silicon generation, card bandwidth and software service are different levels of the product.

DPU generation	Disclosure and product period	Processing and memory	Interfaces and architectural change
Capri, first generation	First silicon in 2018; detailed at Hot Chips 2020	16 nm; 112 MPUs; four Cortex-A72 cores; 4-8 GB HBM	Up to 2 × 100 Gb/s; PCIe 4.0 ×16; establishes the programmable services architecture
Elba, second generation	First silicon in 2020; deployed commercially by 2022	7 nm; 144 MPUs; sixteen Cortex-A72 cores; dual DDR4-3200 interfaces	Up to 2 × 200 Gb/s; 32 PCIe 4.0 lanes; larger service state and host connectivity
Giglio, second generation plus	Announced in 2023; detailed product brief in 2024	Elba-compatible architecture; 144 MPUs; sixteen Cortex-A72 cores; dual DDR5-5600 interfaces	Up to 2 × 200 Gb/s; 16 PCIe 4.0 lanes; optimized power/performance and memory subsystem
Salina, third generation	Announced at Advancing AI 2024; 2025-2026 platform generation	5 nm; programmable P4 services; published card configuration has 64 GB DDR5-6400 with ECC	2 × 400 Gb/s port capability; card service rating 400G; PCIe 5.0 ×16 for a single host; AI front-end and storage-service expansion

The dates distinguish first silicon from later product documentation. The interface figures describe published chip or card capabilities; they do not imply that every board exposes every lane or that all traffic traverses the host interface. In particular, network-to-network processing can operate independently of host PCIe throughput.

4.2 Capri: establishing a programmable services processor

Capri's notable implementation choice was to combine a 16 nm packet processor with HBM in a 2.5D assembly. Its on-package memory served packet-processing state and services rather than the large numerical arrays associated with Instinct. The Hot Chips 2020 presentation disclosed Capri alongside its successor, making the memory-system change between the two unusually easy to follow.

The common architecture separates a packet-processing path from a supporting SoC. Parsers construct packet-header vectors; table lookups and match processing units apply actions; deparsers produce outgoing packets. A network-on-chip connects these pipelines with Arm cores, memory, PCIe and offload engines. This puts programmable packet work close to both the network and service state.

The central design objective is to keep common traffic processing in the fast path. General-purpose Arm cores provide software flexibility for control and exceptional cases; they are not the sole source of

per-packet throughput. AMD's software description extends P4 beyond header rewriting to timers, asynchronous events, notifications and protocol state machines. That allows services such as connection handling and storage transport to exploit the programmable pipeline.

4.3 Elba: more state, more cores and a different memory balance

Elba moved the design to 7 nm and expanded the surrounding SoC. AMD's later product brief specifies 144 MPUs at 2 GHz, sixteen Cortex-A72 cores at 3 GHz, dual DDR4-3200 interfaces supporting 8-64 GB, and 32 PCIe 4.0 lanes. These are later published specifications, distinct from the earlier operating points in the 2020 conference slides.

The move from Capri's HBM to external DDR changes the balance between packaging, memory capacity and service-state storage. The architectural implication is that a DPU needs enough state for connections, mappings and queues, alongside sufficient lookup throughput. HBM is one way to address that problem, but its presence is not the defining property of a DPU.

Elba's dedicated engines accelerate cryptography, compression, checksums, deduplication and erasure coding. Source compatibility with Capri preserves investment in P4 services. PCIe root-complex and endpoint modes, lane partitioning and multihost support allow the same architecture to participate in server cards and network or storage appliances.

VMware's vSphere 8 announcement in August 2022 is an important productization milestone. Pensando DSCs supported the vSphere Distributed Services Engine approach, which moves infrastructure work away from the application's host CPU and separates infrastructure execution from tenant workloads. This is a systems benefit whose value depends on the complete software deployment, not merely DPU core count.

4.4 Giglio: an optimized branch rather than a new programming model

AMD describes Giglio as **second generation plus**. It retains the 144-MPU and sixteen-Cortex-A72 organization and source compatibility with Capri and Elba. Its published memory interface moves to DDR5-5600, while the PCIe interface has sixteen Gen4 lanes. It is therefore more accurate to describe Giglio as a revised balance of resources and efficiency than as a uniform doubling of Elba.

The product remains aimed at stateful networking, security and storage services, with deployments ranging from compact server cards to appliances and smart switches. Its significance is continuity: customers can retain the service model and much of the software while changing the implementation underneath it. This is the DPU counterpart to the platform continuity visible across several EPYC generations, although the silicon organizations are very different.

4.5 Salina: the AI front end becomes a first-class workload

Salina lists two 400 Gb/s port configurations, while the same brief rates the card and service data path at 400G. The port list should therefore not be converted into an asserted 800G service-throughput result. The card brief specifies PCIe 5.0 $\times 16$ for a single host, support for two-host configurations, and 64 GB DDR5-6400 with ECC. Its services include overlays, stateful security, telemetry, IPsec, storage virtualization and NVMe over TCP or RDMA. The progression is from general cloud infrastructure toward the front end of AI systems, where ingestion, tenant isolation and storage access compete with application work for host resources.

The word "front end" describes a role, not an absolute restriction on the silicon. Salina can implement multiple network and storage services; the architectural distinction is that it manages infrastructure around the accelerators. Pollara and Vulcano concentrate on moving data for accelerator communication.

AMD's 2026 networking disclosures extend Salina's role to DPU-managed NVMe storage for AI key-value cache. This adds a memory tier for retained inference state. It does not turn NVMe into HBM: the software must decide what to retain, place and fetch, and the storage/network path has different latency and bandwidth from local accelerator memory. The useful system question is how much expensive GPU memory can be reserved for active computation without making state retrieval the next bottleneck.

4.6 Software and deployment are part of the DPU architecture

Pensando's Software-in-Silicon Development Kit, or SSDK, combines a P4 compiler, Arm toolchain, simulator, libraries, drivers and reference pipelines. P4 targets the fast path; C and C++ target the Arm complex. The environment supports development on x86 systems before deployment to the DPU. Its scope includes data, control and management functions, not just packet filters.

AMD's Policy and Services Manager, or PSM, supplies centralized policy and management for full-stack deployments. Cloud providers can instead integrate their own controllers and service logic. These are alternative software arrangements around the same underlying processing capabilities.

Three deployment forms explain the breadth of the product line:

Deployment form	Representative implementation	What changes in the system
Host-attached services	Pensando DSC; vSphere Distributed Services Engine integration	Infrastructure processing gains a separate execution and management location
Shared network-function pools	Microsoft and AMD Pensando's Sirius system, NSDI 2023	Stateful functions can be provisioned independently of each tenant server
Services embedded in a switch	HPE Aruba Networking CX 10000; Cisco N9300 Smart Switches	Security and telemetry execute near the network path, with a switch ASIC still handling switching

The NSDI 2023 paper Disaggregating Stateful Network Functions is the most direct academic systems companion to the DPU product history. Its Sirius design uses pools of P4-programmable cards, replicates state between paired devices and moves load across the pool. The paper identifies DSC-200 hardware with substantial external memory and integrated Arm cores. It explains how a DPU can support cloud services without requiring an identical services processor beside every application host.

HPE's CX 10000 and Cisco's N9300 show another boundary choice: integrate Pensando processing into the switch. Cisco's 2025 disclosure pairs its Silicon One E100 with AMD Pensando DPUs. These are partner switch products containing AMD technology; their switching capacity should not be attributed to a standalone AMD switch ASIC.

4.7 Pollara 400: programmable transport for GPU clusters

Pollara was announced in October 2024 and made available for purchase in April 2025. Its architectural disclosure at Hot Chips 2025 is particularly useful because it explains the processing machinery behind the AI NIC label.

The product provides up to 400 Gb/s Ethernet and PCIe 5.0 ×16, in half-height, half-length PCIe and OCP 3.0 TSFF card formats. Its network interface can be configured as one 400 Gb/s, two 200 Gb/s or four 100 Gb/s ports. Pollara supports conventional host networking as well as GPU scale-out, including RoCEv2 interoperability and an RCCL-oriented data path.

Kevin Chu's Hot Chips presentation separates network-facing P4NET processing from P4DMA engines that bridge packets and memory. Its match processing units execute a domain-specific instruction set; table engines access the state needed by the transport. The talk also discloses address translation for registered memory, atomic state operations and a broadcast invalidate/update mechanism for pipeline caches. These are important differences from a diagram that simply labels a box “programmable

Ethernet.” They explain how the NIC handles memory access and shared protocol state while many transfers are active.

The principal transport features address different failure modes. Packet spraying spreads traffic across paths. Out-of-order reception allows those paths to have different delays while preserving correct message delivery. Selective retransmission recovers missing data without resending everything that followed it. Path-aware congestion control adjusts behavior when a route is overloaded. These mechanisms determine useful communication throughput, particularly when synchronized GPUs create bursts or incast traffic.

AMD originally described Pollara as “UEC-ready.” That wording identifies a programmable implementation designed for evolving Ultra Ethernet capabilities; it should not be rewritten as unconditional conformance to every later consortium specification. The deployed transport mode, firmware and network configuration remain part of the system definition.

4.8 Vulcano 800: from an adapter to a rack component

Vulcano doubles nominal per-NIC Ethernet bandwidth to 800 Gb/s and appears in AMD's 2026 Helios portfolio. Its current product brief supports RoCEv2, UEC-ready RDMA, Multipath Reliable Connection, or MRC, and custom transports. Programmability allows the operator to choose how much path selection belongs at the NIC or in the switches.

The product has both standard adapter forms and custom multi-NIC boards. The latter are consequential: Helios can allocate two or three Vulcano NICs per GPU. The advertised maximum of 2.4 Tb/s per GPU is therefore a system configuration of three 800 Gb/s NICs. A six-NIC board supplies 4.8 Tb/s in aggregate for two GPUs. These are different quantities from the bandwidth of one NIC or one external connector.

AI NIC configuration	Ethernet bandwidth	Physical or system form	Main generational implication
Pollara 400	Up to 400 Gb/s per NIC	PCIe HHHL or OCP 3.0 TSFF	Programmable AI transport reaches a standard server-adapter platform
Vulcano 800 adapter	Up to 800 Gb/s per NIC	PCIe HHHL or OCP 3.0 TSFF	Higher-bandwidth scale-out with programmable transport choices
Vulcano four-NIC board	3.2 Tb/s per board; 1.6 Tb/s per GPU	Custom Helios board serving two GPUs	Two NICs per GPU
Vulcano six-NIC board	4.8 Tb/s per board; 2.4 Tb/s per GPU	Custom Helios board serving two GPUs	Three NICs per GPU

AMD also describes Vulcano's role in scale-across networking, extending communication between clusters or sites. This makes path health, observability and recovery central product concerns. Additional link bandwidth can shorten transfer time, but geographic distance remains a latency constraint; the placement of training and inference work must account for it.

The progression from Pollara to Vulcano therefore has two dimensions: higher per-device bandwidth and a tighter relationship between NIC count, GPU bandwidth demand and rack design. Treating Vulcano only as an “800G card” misses the second change.

4.9 Alveo SmartNICs and network accelerators

The Xilinx-derived SN1000 family takes a different route to programmability. The SN1022 combines a 16 nm UltraScale+ FPGA with a sixteen-core Arm Cortex-A72 subsystem. P4, high-level synthesis and RTL can customize offloads; the published design includes network virtualization, security and storage functions. Its brief's 200 Gb/s full-duplex throughput figure combines directions around a 100 Gb/s service rate, so it should not be equated with Pensando's dual-200 Gb/s interface capability.

AMD currently supports existing SN1022 users while directing new designs toward Alveo U45N. U45N exposes two 100 Gb/s network interfaces and an FPGA platform for custom infrastructure processing, with Vivado and an OpenNIC reference design. This branch is useful when the customer wants to implement a data path in programmable logic rather than select or extend a Pensando service pipeline.

Hot Chips 2022's AMD 400G Adaptive SmartNIC SoC adds a further architectural disclosure. It integrates hardened networking, a processing subsystem, programmable logic, memory and a network-on-chip. The presentation describes PCIe 5.0/CXL 2.0 connectivity, 2×200 Gb/s or 4×100 Gb/s networking, DDR5/LPDDR5 controllers and cryptographic protection. It belongs in the technology history, but its conference disclosure should not be relabeled as a Pollara or Salina launch.

4.10 Solarflare X2, Alveo X3 and Solarflare X4

The Solarflare-derived adapter line addresses latency-sensitive host applications, supported by Onload and related software. AMD's migration documentation explicitly connects older Solarflare families and X2 to Alveo X3, and subsequently to Solarflare X4. This is a separate ancestry from Pensando.

X4 includes the X4522 and X4542, with PCIe 5.0 $\times 8$, compact passively cooled cards and network configurations extending to dual 100 Gb/s. The ASIC supplies express and enterprise data paths for different application requirements. These products help complete the networking portfolio map, particularly for ingress services and latency-sensitive applications. They should not be presented as replacements for the high-bandwidth GPU collective network.

4.11 Card resources and public disclosure gaps

SN1022 and U45N both combine an UltraScale+ FPGA with sixteen discrete Cortex-A72 cores, 12 GB total DDR4 and a 75 W board limit. Their host interfaces differ: SN1022 lists Gen4 $\times 8$ or Gen3 $\times 16$; U45N lists Gen3 $\times 16$ or two Gen4 $\times 8$ interfaces. AMD supports existing SN1022 users while directing new designs to U45N.

X3522 is a turnkey FPGA-based low-latency NIC; X3522PV exposes the programmable branch. The brief lists four 10/25G ports, 8 GB DDR4 and 35 W in NIC mode, versus 75 W for PV. Solarflare X4 uses a custom ASIC: X4522 has two ports up to 50G; X4542 has two up to 100G or four lower-rate ports. Its PCIe 5 $\times 8$ design is rated below 25 W.

The Hot Chips 2022 adaptive SmartNIC SoC separately discloses sixteen Cortex-A78AE application cores, four Cortex-R52 real-time cores, programmable logic, DDR5/LPDDR5 controllers and PCIe 5/CXL 2.0. No reviewed primary source maps it directly to a Pensando retail generation.

Newer Pensando briefs disclose external interfaces and services more fully than internal pipeline stages, table capacities, embedded cores or power envelopes. Those missing fields remain n/d. Programmable routing, selective retransmission and out-of-order placement still require evaluation under the intended topology and traffic pattern.

Source map: H1-H14; I1-I10; E9, E13; K2, K3; N34, N44, N49. Full references appear in the bibliography; numerical cells are indexed in the companion claims ledger.

5. Adaptive and adjacent accelerators

5.1 Alveo: several accelerator branches under one card brand

The earlier Alveo U200/U250, U280, U50 and U55C generation established FPGA cards for data-center acceleration. Their resources and memory configurations serve different balances of compute, memory and networking; they are not successive names for one fixed AI architecture. AMD's current general-compute portfolio highlights U50, U55C and V80, while older devices remain relevant to the historical software and deployment record.

V80, introduced in 2024, is an instructive comparison with Instinct. It uses a Versal HBM device with 32 GB HBM2e, approximately 810 GB/s peak HBM bandwidth and four 200 Gb/s network connections. Its FPGA fabric supports custom data types and data movement. The supplied Alveo Versal Example Design and Vivado flow target hardware developers. The architectural opportunity is to build a tailored processing pipeline around memory and I/O rather than express all work as GPU kernels. V80's HBM and network ports make it relevant to infrastructure even when it is not doing neural-network matrix multiplication.

V70, announced at CES 2023, is a distinct AI Engine inference branch. AMD lists second-generation AIE-ML tiles, a 75 W card and a small form factor. Its internal-memory bandwidth and external-DDR bandwidth refer to different levels of the memory system and cannot be compared directly with Instinct's HBM bandwidth. As of the report cutoff, AMD marks V70 discontinued, with no further updates to its targeted solutions and limited support. It remains important to the architectural history but is not a current purchasing recommendation.

MA35D, introduced in 2023, demonstrates that Alveo is not exclusively an FPGA brand. The card uses two purpose-built 5 nm video-processing ASICs with codec engines and AI-assisted video-quality processing. It follows the earlier U30 media line and supports AV1 alongside H.264 and H.265, with the AMD Media Acceleration SDK connecting it to frameworks such as FFmpeg and GStreamer. In a video-oriented AI service, dedicated media processing can prepare or deliver streams around an inference pipeline without consuming the same resources as the model itself.

AMD also maintains Alveo products aimed at specialized low-latency workloads, including X3522PV, UL3422, UL3524 and the VCK5000 development platform. These are adjacent infrastructure tools, particularly when a customer combines streaming data, networking and application-specific inference. They are not additional Instinct generations.

5.2 Versal: the silicon architecture behind adaptive acceleration

Versal combines programmable logic, embedded scalar processing, hardened I/O and a programmable network-on-chip. Selected families add AI Engine arrays; the presence and type of AI Engines depend on the series. This distinction matters because a Versal HBM accelerator and a Versal AI Core device need not offer the same execution resources.

Hot Chips 2019's Versal/AI Engine presentation disclosed the VC1902 architecture, including an array of 400 AI Engine tiles with local memories, SIMD execution and intertile communication. The dataflow organization lets neighboring engines exchange intermediate results without routing every operation through external memory. ISSCC 2020's A Versatile 7nm Adaptive Compute Acceleration Platform Processor supplies the complementary silicon-implementation disclosure. This is another useful Hot Chips-ISSCC pairing, alongside those for Zen and Instinct.

Hot Chips 2020 covered Versal Premium, and Hot Chips 2021 covered Versal AI Edge. These extend the architecture toward different priorities: connectivity and hardened infrastructure functions on one branch, and efficient embedded AI on another. The resulting family tree should be read by resources and workload role, not as a single sequence of increasingly large neural accelerators.

Versal branch	Main resource emphasis	Relevance to infrastructure
Prime / Prime Gen 2	Programmable logic, embedded processors and connectivity	General embedded acceleration and custom control/data paths
AI Core / AI Edge / AI Edge Gen 2	AI Engine-based processing alongside logic and processors	Inference, signal processing and sensor-oriented AI
Premium / Premium Gen 2	High-speed I/O, hardened protocol and security resources	Networking, storage, memory expansion and high-bandwidth acceleration
HBM	Programmable processing with nearby high-bandwidth memory	Memory-intensive acceleration; the basis of Alveo V80
RF	Integrated RF-oriented resources and adaptive processing	Radio and sensing infrastructure at the edge

Premium Gen 2 adds PCIe 6.0 and CXL 3.1, faster external-memory interfaces and hardened security functions. AMD reported standard devices shipping in June 2026. The separately announced Memory on Package variant integrates up to 32 GB LPDDR5X and 288 GB/s; its schedule calls for sampling at the end of 2026 and production in the second half of 2027. It is a roadmap extension at this cutoff, not an already completed deployment generation.

The portfolio demonstrates three distinct uses of nearby memory: HBM feeds Instinct's parallel numerical execution; HBM or DDR can hold DPU service state; and Versal's local memories, HBM or packaged LPDDR serve different adaptive-processing designs. The memory technology alone does not identify the processor's role.

5.3 The local and embedded branches

Radeon AI PRO R9700, announced at Computex 2025, brings RDNA 4 and 32 GB GDDR6 to local AI development and inference. Threadripper supplies workstation CPU capacity and I/O, while Ryzen AI and Ryzen AI Halo address compact local systems. These products connect development and smaller deployments to the broader AMD software ecosystem, but their memory, interconnect and deployment scale differ from Instinct systems.

EPYC Embedded extends server-class compute into network, storage and industrial appliances with long product lifecycles. The family spans Embedded 3000, 7002/7003, 8004/9004, and the Zen 5-era 2005, 4005 and 9005 branches. The smaller BGA and AM5 products serve different power and platform requirements from SP5 server-class devices. These CPUs may run the control and storage services around an AI installation without being its model accelerator.

At the physical-AI end, the portfolio includes Zynq UltraScale+ platforms, Versal adaptive SoCs and Kria modules. The older K26/K24 module families and the 2026 Kria AI SOM have different underlying architectures. The newer AI SOM uses Ryzen AI Embedded X100, bringing Zen 5 CPU cores, RDNA 3.5 graphics, an XDNA 2 NPU and unified memory to a COM-HPC module. It extends the product map to robotics and autonomous systems rather than adding another data-center GPU.

Source map: J1-J13; G5, G7, G16, G17; K8. Full references appear in the bibliography; numerical cells are indexed in the companion claims ledger.

6. Systems: nodes, baseboards and racks

6.1 The node is a separate architectural unit

Frontier combines one custom EPYC with four MI250X modules and four 200 Gb/s Slingshot network interfaces. There are eight logical GPU devices because each module contains two GCDs. Dividing node network bandwidth by four physical modules yields 25 GB/s per module per direction; dividing by eight logical GPUs produces a different number. Both calculations need their denominator stated.

El Capitan replaces the separate host-plus-GPU pattern with four MI300A packages. Each integrates twenty-four CPU cores and coherent package HBM. The node therefore has ninety-six CPU cores, four GPU/APU packages and 512 GB of memory distributed across those packages, with four 200 Gb/s Slingshot interfaces. CPU/GPU locality improves inside each APU; communication between APUs remains a system-level path.

The MI300X and MI350 universal-baseboard model uses eight discrete GPU modules with external CPU hosts. OEMs choose CPU count, PCIe switching, NIC ratios, chassis and cooling. A rack containing sixty-four or 128 GPUs does not thereby become one scale-up domain: the direct baseboard domain remains eight unless another fabric is explicitly provided. GPU count per rack, GPU count per node and tightly coupled domain size must be recorded separately.

EPYC also hosts competing accelerators. NVIDIA's DGX A100, introduced at GTC 2020, used two EPYC 7742 CPUs. That is an AMD CPU deployment in a NVIDIA system, not an AMD GPU architecture disclosure. Hyperscaler instance launches similarly establish one service configuration rather than the full capability of a merchant CPU family.

6.2 System-design ownership and ecosystem boundaries

Lineage	Transaction milestone	What is in scope	Sources
Solarflare → Xilinx → AMD	Xilinx acquired Solarflare in July 2019	Low-latency NIC and Onload ancestry, including pre-AMD generations	N44 I9 I10
Xilinx → AMD	AMD completed acquisition 2022-02-14	UltraScale+, Versal, Alveo, Zynq, Kria and adaptive tools	J13
Pensando → AMD	AMD completed acquisition 2022-05-26	Pre-acquisition Capri/Elba and AMD-era DPU/AI NIC successors	H1 H2
Nod.ai / Silo AI → AMD	Nod.ai completed Oct 2023; Silo AI completed Aug 2024	Compiler/software expertise and model development/deployment capability	N51 N52
ZT Systems → AMD / Sanmina	AMD close 2025-03-31; manufacturing sale closed 2025-10-27	AMD retained system design; Sanmina took manufacturing operations	N43 N21

The ZT transactions expanded AMD's system-design capability while leaving manufacturing with a partner. This matters to the scope of Helios: architecture, validation and customer integration are part of AMD's product effort, but not every rack component or manufacturing operation becomes an AMD silicon product. The same distinction applies to HPE, Cisco, Broadcom and other networking partners.

6.3 Helios and the three network roles

Helios brings EPYC, Instinct and Pensando into a rack design. Its significance is the co-design of compute, memory, interconnect, power, cooling and operations. The CPU/GPU generation details appear in Chapters 1 and 2; the infrastructure view explains how the product families cooperate.

Network role	Main traffic	AMD components and integration
Front end and storage access	Requests, ingestion, tenant services, storage and retained inference state	Salina DPU, host networking and storage software
Scale-up within the accelerator domain	Frequent tightly coupled GPU communication	Infinity Fabric in earlier platforms; UALoE in Helios
Scale-out beyond the domain	Distributed training and inference across systems	Pollara or Vulcano AI NICs and the Ethernet fabric

AMD's detailed Helios description uses a single-hop, multiplane UALoE fabric connecting seventy-two GPUs through twelve switches in six switch trays. This is distinct from the Vulcano scale-out network. Combining their bandwidth figures into one undifferentiated “network bandwidth” number would obscure which traffic each fabric carries.

AMD Fabric Manager and Fabric OS provide the scale-up operational layer, including telemetry and recovery; vPods partition the accelerator domain into software-defined environments. ROCm and collective-communication software operate above the physical connectivity. These are necessary parts of turning connected hardware into a usable shared system.

The switch platforms are also an ecosystem story. AMD's HPE collaboration and joint Broadcom session at Advancing AI 2026 describe UALoE switch integration. Helios should therefore be understood as an AMD platform design implemented with system and networking partners, not as evidence that every switch chip, optical module, memory device or cooling component is an AMD product.

6.4 Power and cooling at the correct boundary

Eight MI350X modules sum to 8 kW of rated GPU board power; eight MI355X modules sum to 11.2 kW. CPUs, NICs, switches, fans or pumps, conversion losses and other components add to those figures. They are neither wall-power measurements nor complete rack ratings. Air versus liquid is a platform design choice tied to heat flux, coolant conditions, rack density and serviceability, rather than a label that determines efficiency by itself.

The reviewed final Helios blueprint establishes a liquid-cooled design and its compute/network population, but does not supply a complete rack-power specification suitable for this comparison. The table leaves that field n/d instead of back-solving it from an assumed GPU wattage. Facility planning requires the OEM's final power, cooling and redundancy envelope.

Source map: D8, E3, E7, E8, E13, E17, F8, F9, G6, G14, H12, H13, K3, K4; N14, N15, N21, N41-N44. Full references appear in the bibliography; numerical cells are indexed in the companion claims ledger.

7. Software and hardware enablement

7.1 Software product lines complete the hardware picture

Software or platform	Main hardware relationship	What it contributes
ROCm, HIP and optimized libraries	Instinct and supported Radeon platforms	GPU programming, runtimes, math, AI frameworks and profiling
RCCL and AI NIC integration	Multiple GPUs and scale-out adapters	Collective communication and transport integration
AMD Enterprise AI Reference Stack	AMD compute platforms	Kubernetes-based deployment, inference microservices, AI Workbench, resource management and solution blueprints
ROCm.AI and ROCm Hyperloom	AI development and optimization on supported AMD systems	Integrated workflows and profiling-driven application/kernel optimization
Pensando SSDK and PSM	Pensando DPUs	Service development, policy and fleet integration
Fabric Manager and Fabric OS	Helios scale-up fabric	Provisioning, observability and recovery
Vivado, Vitis, Vitis AI and XRT	FPGAs, Versal and supported Alveo platforms	Hardware design, acceleration, inference compilation and runtime support
Onload and related network software	Solarflare / Alveo Ethernet adapters	Low-latency host networking
Zen Software Studio / ZenDNN	EPYC and other supported Zen processors	CPU libraries, tools and inference optimization
Ryzen AI and robotics software	Ryzen AI, embedded and Kria platforms	Local inference and physical-AI application enablement

The software history follows the hardware's expanding scope. ROCm began as a GPU-compute platform and grew with Instinct into a larger AI and HPC environment. By August 2026, AMD's ROCm 10 announcement emphasized ROCm.AI workflows, Hyperloom optimization and a preview unified CLI. The enterprise reference stack addresses a different layer: deploying and managing applications and resources above the compiler and runtime.

Adaptive-computing software remains a distinct development path. An FPGA design, a Versal AI Engine program, a Pensando P4 service and a CDNA GPU kernel are different artifacts even if they participate in one AI service. Historical Vitis AI documentation also uses “DPU” for **deep-learning processing unit IP**; that term must not be confused with a Pensando **data processing unit**.

7.2 Align software versions with hardware capabilities

The release timeline in the fixed comparison tables distinguishes product announcements from software release evidence. ROCm 6 accompanied MI300's launch in December 2023. MI350 enablement appears in the 6.4.x transition and subsequent compatibility records. ROCm 7 was previewed at Advancing AI 2025, while AMD's release article is dated 16 September. ROCm 10 and its developer-workflow additions were announced on 27 August 2026.

AMD also expanded the software organization through acquisitions: Nod.ai closed in October 2023, adding open-source compiler and deployment expertise; the Silo AI acquisition completed in August

2024, strengthening model development and deployment on AMD hardware. These are engineering capabilities around the platform, not additional processor architectures.

A datatype being present in the ISA is only the first step. The compiler must expose it, libraries must implement useful operations, frameworks must select those operations, and model accuracy must tolerate the numerical format. The same is true of partitioning: supported hardware modes can differ from the combinations accepted by a particular driver, runtime, virtualization stack or collective library. The versioned compatibility matrix remains the authority for a deployment.

RCCL performance depends on both the GPU topology and the network plugin/transport path. More NIC bandwidth cannot hide arbitrary host copies, synchronization or poor collective scheduling. HIP portability can preserve a programming interface while still requiring target-specific kernels, layouts, occupancy choices and communication tuning. The ASPLOS work on compute/collective overlap is useful research context here, but it is not evidence that its proposed hardware exists in a current Instinct product.

The accompanying tools have different compilation targets. Pensando uses P4 for fast-path services and C/C++ on its embedded CPU complex. Vitis and Vivado produce adaptive hardware and AI Engine programs; XRT manages supported accelerator execution. Onload accelerates host networking through a separate driver and user-space stack. A complete AMD installation can use all of these without having one universal compiler or runtime for the entire rack.

Source map: K5-K9; H7, H8; F10; N17-N19, N24, N28, N36, N51, N52. Full references appear in the bibliography; numerical cells are indexed in the companion claims ledger.

8. Fixed-schema comparison tables

These tables preserve the common schemas for use alongside reports on other vendors. The downloadable CSVs contain the same cells without page-width constraints. “Max” refers to the stated family maximum or named reference SKU. A combination of column maxima is not a claim that every SKU exposes all of them simultaneously.

First disclosure and GA are separate in Section 8.7. Historical shipping rows record an established commercial generation; they do not assert continued orderability. Newly launched products with no established broad GA remain limited availability. Roadmap entries have no inferred specifications.

8.1 Host CPUs

Generation	Codename	GA	Status	Core μ arch	Max cores/threads	Process (per die)	Chiplets and packaging	L2 per core	L3 (total; per sharing domain)	Memory (type, channels, speed, peak GB/s)	Sockets and socket links	PCIe/CXL	Max TDP	Matrix/vector ISA
EPYC 7001	Naples	2017-06-20	Shipping (historical GA)	Zen	32/64	GF 14nm SoC	4 Zeppelin SoCs; organic MCM	512 KB	64 MB; 8 MB/4 cores	DDR4; 8 ch; 2667 MT/s; 170.7	1P/2P; 4 x16 IF links in 2P	PCIe 3 x128 (1P); CXL n/a	200 W cTDP	AVX2; 128-bit FP datapath
EPYC 7002	Rome	2019-08-07	Shipping (historical GA)	Zen 2	64/128	TSMC N7 CCD; GF 14nm IOD	Up to 8 CCD + 1 IOD; organic MCM	512 KB	256 MB; 16 MB/4 cores	DDR4; 8 ch; 3200; 204.8	1P/2P; 3-4 x16 xGMI, up to 18 GT/s	PCIe 4 x128 (1P); CXL n/a	280 W (7H12)	AVX2; 256-bit FP
EPYC 7003	Milan	2021-03-15	Shipping (historical GA)	Zen 3	64/128	TSMC N7 CCD; GF 14nm IOD	Up to 8 CCD + 1 IOD; MCM	512 KB	256 MB; 32 MB/8 cores	DDR4; 8 ch; 3200; 204.8	1P/2P; 3-4 x16 xGMI	PCIe 4 x128 (1P); CXL n/a	280 W	AVX2; 256-bit FP
EPYC 7003X	Milan-X	2022-03-21	Shipping (historical GA)	Zen 3	64/128	N7 CCD/cache; GF 14nm IOD	8 CCD + 8 cache dies + IOD; hybrid-bonded cache	512 KB	768 MB; 96 MB/8 cores	DDR4; 8 ch; 3200; 204.8	1P/2P; Milan xGMI	PCIe 4 x128 (1P); CXL n/a	280 W	AVX2; 256-bit FP
EPYC 9004	Genoa	2022-11-10	Shipping	Zen 4	96/192	TSMC N5 CCD; N6 IOD	Up to 12 CCD + IOD; organic MCM	1 MB	384 MB; 32 MB/8 cores	DDR5; 12 ch; 4800; 460.8	1P/2P; 3-4 x16 xGMI, up to 32 Gb/s/lane	PCIe 5 x128 (1P); CXL 1.1+, up to x64	400 W cTDP	AVX-512; 256-bit datapath; BF16/VNNI
EPYC 9004X	Genoa-X	2023-06-13	Shipping	Zen 4	96/192	N5 CCD; N6 IOD; cache node n/d here	Up to 12 CCD + 12 cache dies + IOD	1 MB	1152 MB; 96 MB/8 cores	DDR5; 12 ch; 4800; 460.8	1P/2P; Genoa xGMI	PCIe 5 x128; CXL 1.1+	400 W cTDP	AVX-512; 256-bit; BF16/VNNI
EPYC 97x4	Bergamo	2023-06-13	Shipping	Zen 4c	128/256	TSMC N5 CCD; N6 IOD	Up to 8 x16-core CCD + IOD	1 MB	256 MB; 16 MB/8 cores; 2 CCX/CCD	DDR5; 12 ch; 4800; 460.8	1P/2P; Genoa xGMI	PCIe 5 x128; CXL 1.1+	400 W cTDP	Zen 4 ISA; AVX-512/BF16/VNNI
EPYC 8004	Siena	2023-09-18	Shipping	Zen 4c	64/128	TSMC N5 CCD; N6 IOD	Up to 4 x16-core CCD + IOD; SP6	1 MB	128 MB; 16 MB/8 cores	DDR5; 6 ch; 4800; 230.4	1P; socket links n/a	PCIe 5 x96; CXL 1.1+, up to x48	225 W cTDP	AVX-512; 256-bit; BF16/VNNI

Generation	Codename	GA	Status	Core µarch	Max cores/threads	Process (per die)	Chiplets and packaging	L2 per core	L3 (total; per sharing domain)	Memory (type, channels, speed, peak GB/s)	Sockets and socket links	PCIe/CXL	Max TDP	Matrix/vector ISA
EPYC 9005 classic	Turin	2024-10-10	Shipping	Zen 5	128/256	TSMC N4 CCD; N6 IOD	Up to 16 x8-core CCD + IOD; SP5	1 MB	512 MB; 32 MB/8 cores	DDR5; 12 ch; 6400 current; 614.4	1P/2P; 3-4 x16 xGMI up to 32 Gb/s/lane	PCIe 5 x128; CXL 2.0 up to x64	500 W	AVX-512; 512-bit FP; BF16/VNNI
EPYC 9005 dense	Turin Dense	2024-10-10	Shipping	Zen 5c	192/384	TSMC N3 CCD; N6 IOD	Up to 12 x16-core CCD + IOD; SP5	1 MB	384 MB; 32 MB/16 cores	DDR5; 12 ch; 6400 current; 614.4	1P/2P; Turin xGMI	PCIe 5 x128; CXL 2.0 up to x64	500 W	AVX-512; 512-bit server FP; BF16/VNNI
EPYC 4004	Raphael-derived	2024-05-21	Shipping	Zen 4	16/32	TSMC N5 CCD; N6 IOD	Up to 2 CCD + IOD; AM5; PX V-Cache variants	1 MB	Up to 128 MB (PX); 32/96 MB per CCX	DDR5 UDIMM; 2 ch; 5200; 83.2	1P; socket links n/a	PCIe 5 x28 total; CXL n/a	170 W	AVX-512; 256-bit; BF16/VNNI
EPYC 4005	Grado	2025-05-13	Shipping	Zen 5	16/32	TSMC N4 CCD; N6 IOD	Up to 2 CCD + IOD; AM5	1 MB	Up to 128 MB (PX); 32/96 MB per CCX	DDR5 UDIMM; 2 ch; 5600; 89.6	1P; socket links n/a	PCIe 5 x28 total; CXL n/a	170 W	AVX-512; 512-bit; BF16/VNNI
EPYC 8005	Sorano	2026; exact GA n/d	Shipping	Zen 5-based	84/168	Compute node: conflicting 4nm/5nm in guide; IOD n/d	Up to 12 CCD + IOD; 7 enabled cores/CCD on 84-core SKU; SP6	1 MB	384 MB; 32 MB per CCX/CCD	DDR5; 6 ch; 6400; 307.2	1P; socket links n/a	PCIe 5 x96 + x8 Gen3; CXL 2.0 up to x48	225 W	AVX-512; detailed datapath n/d here
EPYC 9006 (9996 reference)	Venice	n/d; launched 2026-07-23	Sampling or limited availability	Zen 6/6c family	256/512	TSMC N2 compute; other die nodes n/d here	SP7 reference; EFB 2.5D; detailed die population n/d	n/d	1024 MB on 9996; domain n/d	DDR5; 16 ch; RDIMM 8000 / MRDIMM 12800; 1024 / 1638.4	1P/2P; link details n/d	PCIe 6 x96 on 9996; CXL details n/d	600 W default CPU power; range 400-600 W	Detailed instruction/datapath disclosure n/d here
Future EPYC host	Verano	n/d; Helios 500 planned 2027	Roadmap	n/d	n/d	2nm roadmap; other dies n/d	LPDDR integration announced; topology n/d	n/d	n/d	LPDDR integration; channels/rate/capacity n/d	n/d	n/d	n/d	n/d

8.2 AI and HPC accelerators

Generation	Codename /arch	GA	Status	Process	Dies and packaging	Compute units	Dense BF16	Dense FP8	Dense FP6/FP4	FP64 (vector/matrix)	HBM (gen, stacks, GB, TB/s)	Last-level cache	Scale-up (links, GB/s per direction, domain)	Host link	TBP and cooling	Form factor
MI6 / MI8	Polaris / Fiji; GCN	2017	Shipping (historical GA)	14nm / 28nm	Monolithic GPU; MI8 2.5D HBM	36 / 64 CU	n/a	n/a	n/a	n/d; matrix n/a	MI6: GDDR5 16 GB, 0.224 TB/s; MI8: HBM1 x4, 4 GB, 0.512 TB/s; heights n/d	2 MB L2 each	n/a; PCIe communication	PCIe 3 x16	150 / 175 W board; passive air	PCIe cards
MI25	Vega 10; GCN5	2017	Shipping (historical GA)	GF 14nm	1 GPU + 2 HBM2; 2.5D	64 CU	n/a	n/a	n/a	0.768 TF/s / n/a	HBM2 x2; height n/d; 16 GB; 0.484 TB/s	4 MB L2	n/a; PCIe communication	PCIe 3 x16	300 W board; passive air	PCIe card
MI50 / MI60	Vega 20; GCN5.1	2018-2019; SKU-specific	Shipping (historical GA)	TSMC N7	1 GPU; 2.5D HBM	60 / 64 CU	n/a	n/a	n/a	6.7 / 7.4 TF/s vector; matrix n/a	HBM2 x4; height n/d; MI50 16/32 GB, MI60 32 GB; 1.024 TB/s	4 MB L2	2 IF links; 50 GB/s/link; 4-GPU hive	PCIe 4 x16	300 W board; passive air	PCIe cards
MI100	Arcturus / CDNA	2020-11	Shipping (historical GA)	TSMC N7	1 GPU + 4 HBM2; 2.5D	120 CU	92.3 TF/s	n/a	n/a	11.5 TF/s / n/a	HBM2 x4; height n/d; 32 GB; 1.229 TB/s	8 MB L2	3 IF links; 46 GB/s/link; 4 GPUs	PCIe 4 x16	300 W TDP; passive air	PCIe card
MI250 / MI250X	Aldebaran / CDNA 2	2021-11; system dependent	Shipping	TSMC N6	2 GCD + 8 HBM2e; EFB	208 / 220 CU	362.1 / 383 TF/s	n/a	n/a	45.3 / 47.9 vector; 90.5 / 95.7 matrix TF/s	HBM2e x8; height n/d; 128 GB; 3.2 TB/s	16 MB L2 total; 8 MB/GCD	8 external IF/package; 50 GB/s/link; node-specific domain	PCIe 4 x16 or coherent IF system	500 W TDP; MI250X 560 W peak; liquid in Frontier	OAM
MI210	Aldebaran / CDNA 2	2022-03-22	Shipping	TSMC N6	1 GCD + 4 HBM2e; 2.5D	104 CU	181 TF/s	n/a	n/a	22.6 / 45.3 TF/s	HBM2e x4; height n/d; 64 GB; 1.6 TB/s	8 MB L2	3 IF links; 50 GB/s/link; platform dependent	PCIe 4 x16	300 W TDP; passive air	PCIe card
MI300A	CDNA 3 + Zen 4	2023-12; system dependent	Shipping	TSMC N5 compute; N6 IOD	6 XCD + 3 CPU CCD on 4 IOD; 3D hybrid bonding	228 CU + 24 CPU cores	980.6 TF/s	1961 TF/s, E4M3/E5M2	n/a	61.3 / 122.6 TF/s	HBM3 x8; 8-high; 128 GB; 5.3 TB/s	256 MB Infinity Cache; separate CPU L3	8 IF capability; 64 GB/s/link; 4 APUs in El Capitan	On-package CPU coherent fabric; PCIe 5 x16 I/O	550 W TDP / 760 W peak; liquid in El Capitan	SH5 socket APU
MI300X	CDNA 3	2023-12-06	Shipping	TSMC N5 XCD; N6 IOD	8 XCD on 4 IOD; hybrid-bonded 3D + HBM	304 CU	1307.4 TF/s	2614.9 TF/s, E4M3/E5M2	n/a	81.7 / 163.4 TF/s	HBM3 x8; 12-high; 192 GB; 5.3 TB/s	256 MB Infinity Cache	7 active peer links; 64 GB/s/link; 448 aggregate; 8 GPUs	PCIe 5 x16; ~63 GB/s/direction line payload ceiling	750 W peak TBP; air or liquid platform	OAM / 8-GPU UBB

Generation	Codename /arch	GA	Status	Process	Dies and packaging	Comput e units	Dense BF16	Dense FP8	Dense FP6/FP4	FP64 (vector/matrix)	HBM (gen, stacks, GB, TB/s)	Last-level cache	Scale-up (links, GB/s per direction, domain)	Host link	TBP and cooling	Form factor
MI325X	CDNA 3	Q4 2024; OEM expansion Q1 2025	Shipping	TSMC N5 XCD; N6 IOD	MI300X chiplet organization; HBM3e refresh	304 CU	1307.4 TF/s	2614.9 TF/s, E4M3/E5M2	n/a	81.7 / 163.4 TF/s	HBM3e x8; height n/d here; 256 GB; 6 TB/s	256 MB Infinity Cache	7 peers; 64 GB/s/link; 448 aggregate; 8 GPUs	PCIe 5 x16	1000 W peak TBP; cooling system dependent	OAM / UBB
MI350X	CDNA 4	2025-06	Shipping	TSMC N3P XCD; N6 IOD	8 XCD on 2 IOD; hybrid bonding	256 CU	2307 TF/s	4614 TF/s; OCP FP8/MXFP8	MXFP6 9227; MXFP4 9227 TF/s	72.1 / 72.1 TF/s	HBM3e x8; 12-high; 288 GB; 8 TB/s	256 MB Infinity Cache	7 peers; 76.8 GB/s/link; 537.6 aggregate; 8 GPUs	PCIe 5 x16	1000 W TBP; air-cooled platform	OAM / UBB
MI355X	CDNA 4	2025-06	Shipping	TSMC N3P XCD; N6 IOD	8 XCD on 2 IOD; hybrid bonding	256 CU	2517 TF/s	5033 TF/s; OCP FP8/MXFP8	MXFP6 10066; MXFP4 10066 TF/s	78.6 / 78.6 TF/s	HBM3e x8; 12-high; 288 GB; 8 TB/s	256 MB Infinity Cache	7 peers; 76.8 GB/s/link; 537.6 aggregate; 8 GPUs	PCIe 5 x16	1400 W TBP; direct liquid cooling	OAM / UBB
MI350P	CDNA 4	2026; exact GA n/d	Shipping	TSMC N3 XCD; N6 IOD	4 XCD on 1 IOD; 4 HBM stacks	128 CU	1150 TF/s	2300 TF/s; OCP FP8/MXFP8	MXFP6 4600; MXFP4 4600 TF/s	36 / 36 TF/s	HBM3e x4; height n/d; 144 GB; 4 TB/s	128 MB Infinity Cache	Dedicated peer link n/d; PCIe multi-GPU systems	PCIe 5 x16	600 W max TBP; 450 W configurable; passive air	Full-height, double-slot PCIe
MI308 regional branch	MI300 family; detailed arch n/d in filing	Shipments resumed end FY2025	Sampling or limited availability	n/d	n/d	n/d	n/d	n/d	n/d	n/d	n/d	n/d	n/d	n/d	n/d	n/d
MI455X	CDNA 5	n/d; launched 2026-07-23	Sampling or limited availability	TSMC N2/N3; per-die node mapping n/d here	8 XCD + 2 fabric/cache dies + 2 IOD	256 WGP (not CDNA4 CU equivalent)	5033 TF/s	20133 TF/s; OCP FP8/MXFP8	MXFP6 20133; MXFP4 40265 TF/s	5 / 5 TF/s	HBM4 x12; height n/d; 432 GB; 23.3 TB/s	192 MB global L2	UALoE; 1800 GB/s aggregate/direction; 72 GPUs	IF CPU-GPU 128 GB/s/direction; external NIC attach disclosed separately	TBP n/d; direct liquid cooling	EAM / Helios
MI430X	CDNA 5 HPC branch	Planned 2027	Announced	n/d	n/d	n/d	n/d	n/d	n/d	Up to 288 TF/s hardware FP64; vector/matrix split n/d	n/d	n/d	n/d	n/d	n/d	n/d
MI500 / MI600	Next CDNA; details n/d	Roadmap 2027 / 2028	Roadmap	n/d	n/d	n/d	n/d	n/d	n/d	n/d	n/d	n/d	n/d	n/d	n/d	Helios 500 / 600 roadmap

8.3 DPUs, AI NICs and network adapters

Product	GA	Status	Class	Ports × speed	SerDes	Host interface	Programmable engine	Embedded CPU	On-card memory	Transport/RDMA	Congestion control	Key offloads	Power
Capri	By 2020; exact GA n/d	Shipping (historical GA)	DPU	2 x100GbE	25G NRZ	PCIe 4 x16	112 P4 MPUs	4 Cortex-A72	4-8 GB HBM	SDN pipeline; RDMA n/d	n/d	Network/storage/security services	<30 W chip, HC2020 condition
Elba	By 2022; exact GA n/d	Shipping	DPU	2 x200GbE	50G PAM4	PCIe 4 x32; multihost	144 P4 MPUs, 2 GHz brief	16 Cortex-A72, 3 GHz	DDR4-3200; 8-64 GB	Programmable SD N/storage/RDMA	n/d	Crypto, compression, storage, virtualization	HC2020 chip <50 W at 200GE; later card n/d
Giglio	2023-2024; exact GA n/d	Shipping	DPU, 2nd gen plus	2 x200GbE	56G PAM4	PCIe 4 x16	144 P4 MPUs at 2 GHz	16 Cortex-A72 at 3 GHz	Dual DDR5-5600; 8-64 GB	Programmable SD N/storage/RDMA	n/d	Network/security/storage, telemetry	n/d
Salina	2025-2026; exact GA n/d	Shipping	DPU	2 x400GbE port capability; service rating 400G	QSFP112; NRZ/PAM4	PCIe 5 x16; up to 2 hosts	P4 services pipeline; count n/d	n/d	64 GB DDR5-6400 ECC	RoCEv1/v2; NVMe/TCP/RDMA; UEC-ready	Programmable; algorithm n/d	SDN, IPsec/TLS, NVMe, security, telemetry	n/d
Pollara 400	2025-04-09	Shipping	AI NIC	1 x400 / 2 x200 / 4 x100GbE	n/d in brief	PCIe 5 x16	P4NET + P4DMA; MPU/table stages	n/d	n/d	RoCEv2; UEC-ready RDMA	Adaptive spray; path-aware congestion avoidance	RDMA, selective retransmission, out-of-order placement	n/d
Vulcano 800	n/d; 2026 introduction	Sampling or limited availability	AI NIC	1 x800 / 2 x400 / 8 x100GbE per adapter	200G SerDes in rack disclosure	PCIe 6 adapter; UALink custom GPU attach	Programmable transport	n/d	n/d	RoCEv2; UEC-ready; MRC/custom	NIC or switch spray; NIC source routing	Selective retransmission, out-of-order placement, RDMA	n/d; air adapter / liquid custom board
Alveo SN1022	2021; exact GA n/d	Cancelled or repositioned	Adaptive SmartNIC	2 x100GbE ports; 100G service rating	n/d	PCIe 4 x8 or Gen3 x16	UltraScale+ FPGA; P4/HLS/RTL	16 Cortex-A72 at 2 GHz	12 GB DDR4 (4 GB Arm + 8 GB FPGA)	Programmable networking; RDMA feature n/d	User-design dependent	virtio, vSwitch, security and storage offload	75 W maximum board
Alveo U45N	n/d	Shipping	FPGA network accelerator	2 x100GbE	n/d	PCIe 3 x16 or 2 x Gen4 x8	UltraScale+ XCU26 FPGA / OpenNIC	Discrete 16 Cortex-A72 at 2 GHz	12 GB DDR4 (4 GB Arm + 8 GB FPGA)	User design / OpenNIC	User design	Programmable virtual switching/storage/security	75 W board; passive air
400G Adaptive SmartNIC SoC	n/d; HC2022 disclosure	Announced	Adaptive SmartNIC SoC	400G aggregate; example 2 x200GbE	112G	PCIe 5 / CXL 2.0	Hardened packet path + programmable logic	16 Cortex-A78AE + 4 Cortex-R52	8 DDR5/LPDDR5 controllers; board capacity n/a	Programmable network interfaces	n/d	Inline cryptography, infrastructure offload	n/d
Solarflare X2	Before 2022; exact GA n/d	Shipping (historical GA)	Low-latency host NIC	X2522 representative: 2 x10/25GbE	n/d	PCIe 3 x8	Fixed ASIC + Onload data path	n/d	n/d	Ethernet/TCP/UDP; kernel bypass	Host stack dependent	PTP, filtering, Onload	n/d
Alveo X3 / X3522	2022; exact GA n/d	Shipping	Low-latency host NIC	4 x10/25GbE	n/d	PCIe 4 x8 or Gen3 x8	UltraScale+ XCUX35 FPGA; turnkey NIC firmware	n/d	8 GB DDR4	Ethernet/TCP/UDP; Onload	Host stack dependent	PTP, kernel bypass	35 W NIC mode; passive air
Solarflare X4	2025-2026; exact GA n/d	Shipping	Low-latency host NIC	X4522: 2 x1/10/25/50; X4542: 2 x40/50/100 or 4 x1/10/25GbE	Up to 53G PAM4	PCIe 5 x8; CXL listed	Express/enterprise ASIC paths	n/d	n/d	Ethernet/TCP/UDP; Onload	Host stack dependent	PTP, timestamping, kernel bypass	<25 W; passive low-profile

8.4 Scale-up interconnect

Name/version	Year	Lane rate	Lanes per link	GB/s per link per direction	Links per device	Topology	Max domain	Coherence/memory semantics
Naples IF socket links	2017	up to 10.7 GT/s (configuration dependent)	16	n/d useful payload	4 external/socket	Direct socket links; distributed memory controllers	2 sockets	CPU cache-coherent NUMA
Rome/Milan xGMI	2019/2021	Rome up to 18 GT/s	16	36 raw at 18 GT/s; payload lower	3-4/socket in 2P	Central IOD per socket; xGMI between IODs	2 sockets	CPU cache-coherent NUMA
Genoa/Turin xGMI	2022/2024	up to 32 Gb/s/lane	16	64 raw; useful payload lower	3-4/socket	IOD-to-IOD	2 sockets	CPU cache-coherent NUMA; NPS controls placement domains
Vega 20 external IF	2018	n/d here	n/d	50	2/GPU	4-GPU hive; platform dependent	4 GPUs	Peer GPU memory access; not uniform local HBM
MI100 external IF	2020	n/d here	n/d	46	3/GPU	4-GPU fully connected hive	4 GPUs	Peer GPU access
MI250X internal IF	2021	25 Gb/s	Aggregate 64 lanes/direction, derived	200 aggregate between GCDs	Internal link group	Direct GCD-to-GCD EFB	2 GCD/package	Two GPU devices; traffic crosses a real locality boundary
MI200 external IF	2021	25 Gb/s	16	50	8/package (4/GCD)	Node-specific mesh incl CPU/GPU attach	Frontier: 4 modules / 8 GPU devices	GPU peer links; coherent CPU connection in supported system
MI300 external IF	2023	32 Gb/s	16	64	8 physical; 7 peer links in MI300X UBB	Direct all-to-all mesh, 8 GPUs	8 GPU UBB; APU system differs	GPU load/store; MI300A adds coherent CPU/shared HBM
MI350 external IF	2025	38.4 Gb/s	16	76.8	7 peer links; host link separate	Direct 8-GPU all-to-all	8 GPUs/UBB	GPU peer memory; compute/memory partitioning configurable
Helios UALoE	2026	n/d electrical rate in reviewed overview	Vendor labels 36 links (x2); physical subdivision not inferred	n/d per physical link; 1800 aggregate/GPU/direction	36 (x2) vendor terminology	Single-hop, multiplane; 12 switches in 6 trays	72 GPUs in Helios	GPU load/store domain; distributed HBM; not asserted CPU cache-coherent rack

8.5 Systems

Platform	Year	CPUs/accelerators	Scale-up domain	Scale-out per accelerator	Rack power	Cooling
EPYC + MI25 server era	2017	EPYC 7001 + PCIe MI25; OEM topology varies	PCIe-attached GPUs; count platform dependent	n/d; partner NIC	n/d	Air, platform dependent
EPYC + MI100 reference era	2020	EPYC 7002 + 4/8 MI100	4-GPU IF hive; 8-GPU nodes may have 2 hives	n/d; partner NIC	n/d	Air, platform dependent
Frontier / HPE Cray EX node	2021-2022	1 custom EPYC + 4 MI250X (8 GPU devices)	4 physical packages / 8 GPU devices	4 x200 Gb/s Slingshot/node; 25 GB/s per package, derived	n/d uniform rack rating	Direct liquid
MI300X universal baseboard	2023	8 MI300X; external host CPU(s) OEM-dependent	8 GPUs; 448 GB/s/direction/GPU	n/d baseline; 1 x400G/GPU example = 50 GB/s	n/d	Air or liquid
El Capitan node	2024	4 MI300A; 96 integrated CPU cores	4 APUs; 512 GB distributed shared-package HBM	4 x200 Gb/s Slingshot/node; 25 GB/s/APU derived	n/d uniform rack rating	Direct liquid
MI350X / MI355X UBB	2025	8 MI350X or MI355X; external host CPU(s)	8 GPUs; 537.6 GB/s/direction/GPU	1 x400G Pollara/GPU example = 50 GB/s; OEM configurable	GPU-only 8 / 11.2 kW per baseboard; full rack n/d	Air (350X) / direct liquid (355X)
Helios reference rack	2026	18 Venice + 72 MI455X; 18 compute trays	72 GPUs; 1800 GB/s/direction/GPU	Up to 3 x800G Vulcano/GPU = 300 GB/s/direction	n/d in reviewed final blueprint	Direct liquid
Helios 500 / 600 roadmap	2027 / 2028 planned	MI500 + Verano / MI600 + Ferrara	n/d	Como/Monza; Palma/Levanzo networking roadmap	n/d	n/d

8.6 Software chronology

Release	Date	Hardware enabled	Key features
ROCm initial platform	2016	Supported GCN GPUs; Radeon Instinct lineage follows	Open heterogeneous runtime/compiler foundations and HIP programming model
ROCm 4.0 era	2020-11 announcement	MI100 / CDNA	Matrix-core compiler/library enablement; BF16 and HPC compute platform
ROCm MI200 enablement	2021-2022	MI250/X and MI210 / CDNA 2	FP64 matrix instructions in rocBLAS; two-GCD placement and coherent systems support
ROCm 6.0	2023-12	MI300A/X / gfx942	MI300 enablement, LLM libraries/frameworks, FP8 paths; software support is datatype/library dependent
ROCm 6.2 era	2024	MI300 series; MI325X generation	LLM performance and library evolution; hardware memory refresh does not imply a new ISA target
ROCm 6.4.x MI350 transition	2025	MI350X/MI355X / gfx950	New architecture and MX formats enter supported compiler, runtime and library combinations
ROCm 7.0	2025-09-16 release article	MI350 and supported other AMD GPUs	Broader MXFP4/MXFP6 support, optimized libraries, framework and developer tooling
ROCm 10	2026-08-27 announcement	MI455X/Helios plus supported AMD portfolio	ROCm.AI developer workflows, Hyperloom, unified CLI preview; version-specific support matrix
Pensando SSDK / PSM	2023-11 brief; maintained subsequently	Capri/Elba/Giglio/Salina as supported	P4 fast path; C/C++ Arm toolchain; simulator; policy and services management
Fabric Manager / Fabric OS	2026	Helios UALoE	Fabric provisioning, telemetry, vPods and fault recovery
Vitis / Vitis AI / XRT	2019 onward; release-specific	Alveo, Versal, adaptive devices	Spatial hardware design, AIE graph compilation, runtime; older DPU IP distinct from Pensando DPU
Onload	Solarflare legacy through 2026	Supported X2/X3/X4 adapters	User-level TCP/UDP acceleration and kernel bypass; NIC/driver/version compatibility matters

8.7 Disclosure and availability chronology

Product	First material disclosure	Commercial / GA milestone	Status at cutoff	Sources
EPYC 7001	Zen at HC2016; EPYC branding May 2017	2017-06-20 launch with partner systems	Shipping; historical GA	A1 G1 N4
EPYC 7002	Next Horizon, 2018-11-06	2019-08-07 launch	Shipping; historical GA	D4 A8
Milan / Milan-X	Zen 3 in 2020; Milan-X preview Nov 2021	Milan 2021-03-15; Milan-X 2022-03-21	Shipping; historical GA	B2 B3 B6 G3
Genoa / Bergamo / Siena	FAD and launch disclosures 2022-2023	Genoa Nov 2022; Bergamo/Genoa-X Jun 2023; Siena Sep 2023	Shipping	G2 B12 B13 B14
Turin	Computex preview, June 2024	2024-10-10 launch and availability	Shipping	G4 C4
Sorano / EPYC 8005	2026 product disclosure	Launch 2026-05-19; exact independent GA day n/d	Shipping; current portfolio	C10 N45 N46
Venice / EPYC 9006	Earlier roadmap; production ramp 2026-05-21	Launch 2026-07-23; broad GA date n/d	Sampling or limited availability	N23 C8 N30
Radeon Instinct MI6/8/25	2016-12 announcement	2017 product generation	Shipping; historical GA	D1 D2
MI50 / MI60	2018-11-06 Next Horizon	Late 2018-2019, SKU-dependent	Shipping; historical GA	D3 D4
MI100	2020-11-16	November 2020 platform introduction	Shipping; historical GA	D6 N5
MI250 / MI250X / MI210	MI200 launch 2021-11-08	MI200 in 2021 systems; MI210 2022-03-22	Shipping; system-specific availability	D9 N6 N7

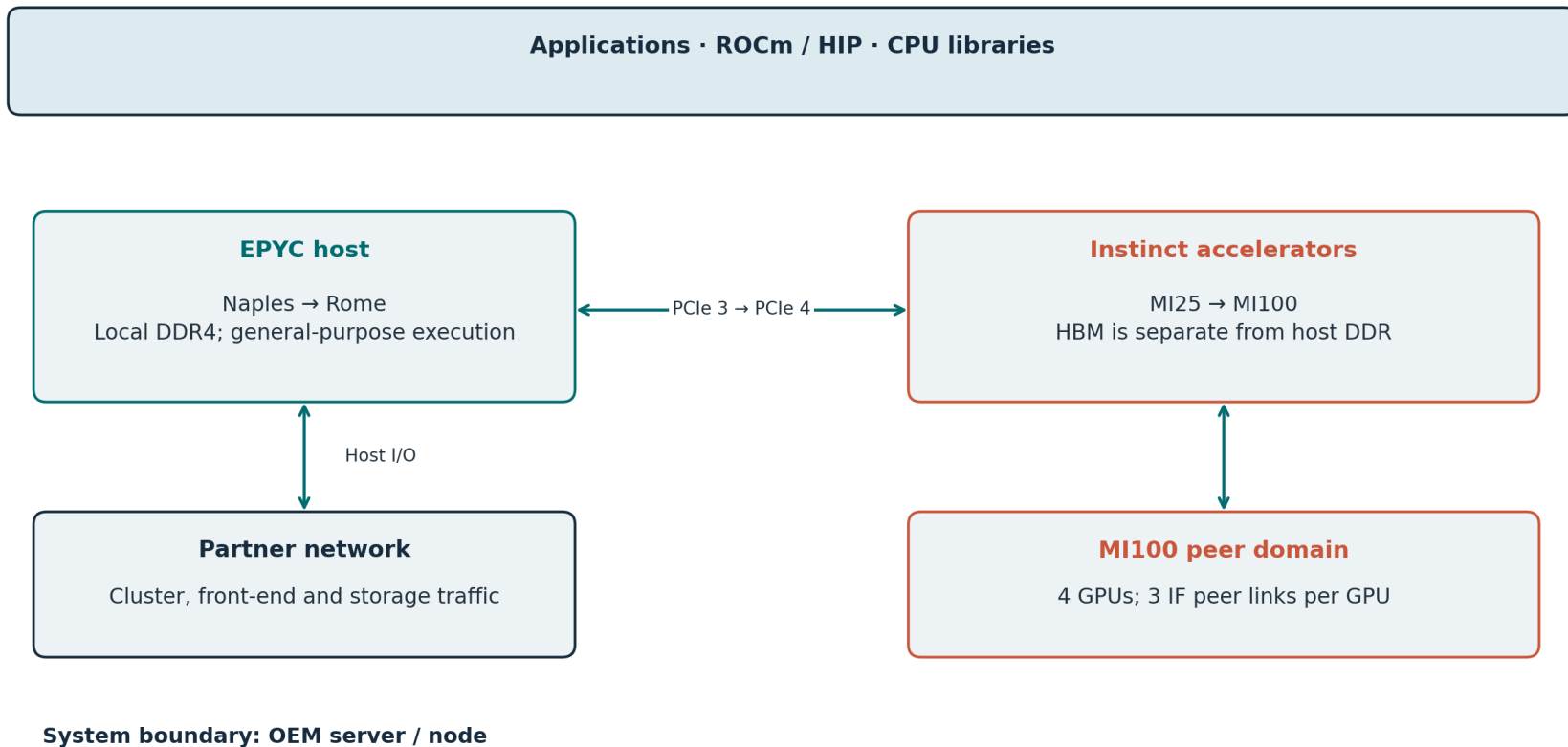
Product	First material disclosure	Commercial / GA milestone	Status at cutoff	Sources
MI300A / MI300X	MI300 CES 2023; MI300X June 2023	2023-12-06 launch; Azure MI300X GA May 2024	Shipping	B13 E6 G12
MI325X	Computex June 2024: 288 GB preview	Final 256 GB Oct 2024; Q4 production, OEM expansion Q1 2025	Shipping	N35 E9
MI350X / MI355X	Roadmap 2024; product details 2025	Advancing AI 2025-06-12	Shipping	N35 E13
MI350P	Detailed AMD blog 2026-05-07	2026 PCIe systems; exact GA day n/d; featured again July	Shipping	N40 N13 C8
MI308	MI300 regional branch; earlier specifications not established here	Licensed shipments resumed end FY2025	Sampling or limited availability, license-dependent	N16
MI455X / Helios	Helios preview Jun 2025; OCP Oct 2025	Production/launch announced Jul 2026; customer rollout varies	Sampling or limited availability	E13 G6 C8 N41
Capri / Elba	First silicon 2018 / 2020; HC2020	Capri historical; Elba deployed commercially by 2022	Shipping; first GA dates n/d	H2 H11
Giglio / Salina	June 2023 / October 2024	2023-2026 portfolio and partner deployments	Shipping; exact GA dates n/d	H5 H6 H13
Pollara 400	Advancing AI, 2024-10-10	Purchase availability 2025-04-09	Shipping	E9 I3
Vulcano 800	Helios roadmap 2025; detailed 2026	2026 introduction; broad adapter GA date n/d	Sampling or limited availability	E13 I4 I5

Per-row primary-source keys, disclosure dates and normalization notes are in row_sources.json and the claims ledger. Accelerator matrix throughput is TFLOP/s throughout; package HBM capacity and bandwidth follow final product specifications. MI250X bandwidth is the vendor-rounded 3.2 TB/s package value. Its detailed interface arithmetic can produce 3.2768 TB/s; derived ratios here consistently use 3.2 rather than mixing the two.

9. Integrated AI-infrastructure stack

2017-2020 | A CPU host and discrete GPU memory

Representative EPYC + Instinct pattern; optional peer fabric depends on GPU generation

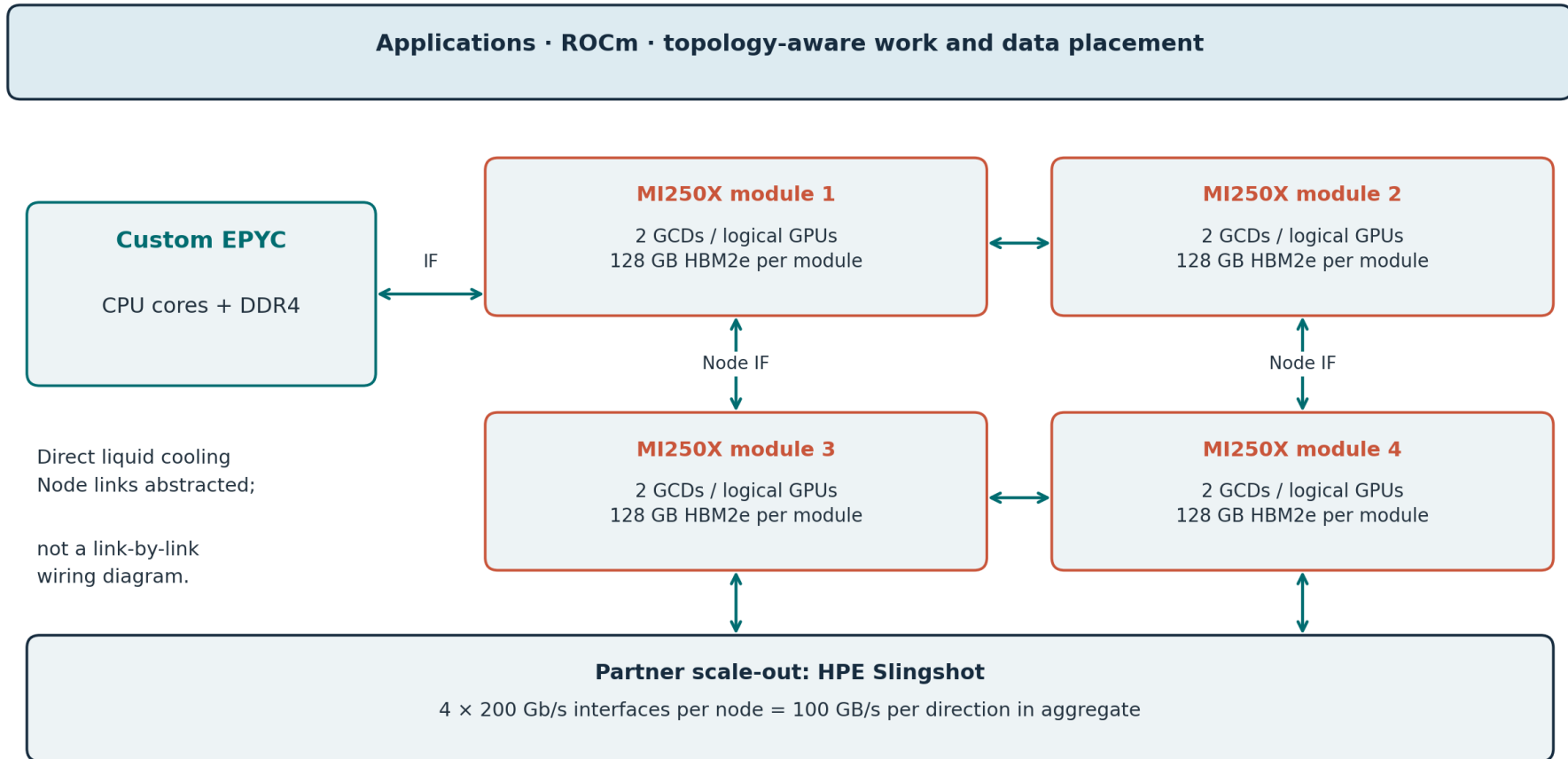


Sources: A2, A8, D1, D5, D6. Topology is schematic; no mandatory GPU or NIC population is implied.

In the first era, EPYC supplies general compute, host memory and PCIe attachment. MI25 and later MI100 supply their own HBM. Partner NICs handle cluster traffic; a four-GPU IF hive is available in the MI100 generation. CPU memory, accelerator memory and the network remain separate resources.

2021-2022 | Frontier combines packages into a node

1 custom EPYC + 4 MI250X modules = 8 logical GPU devices

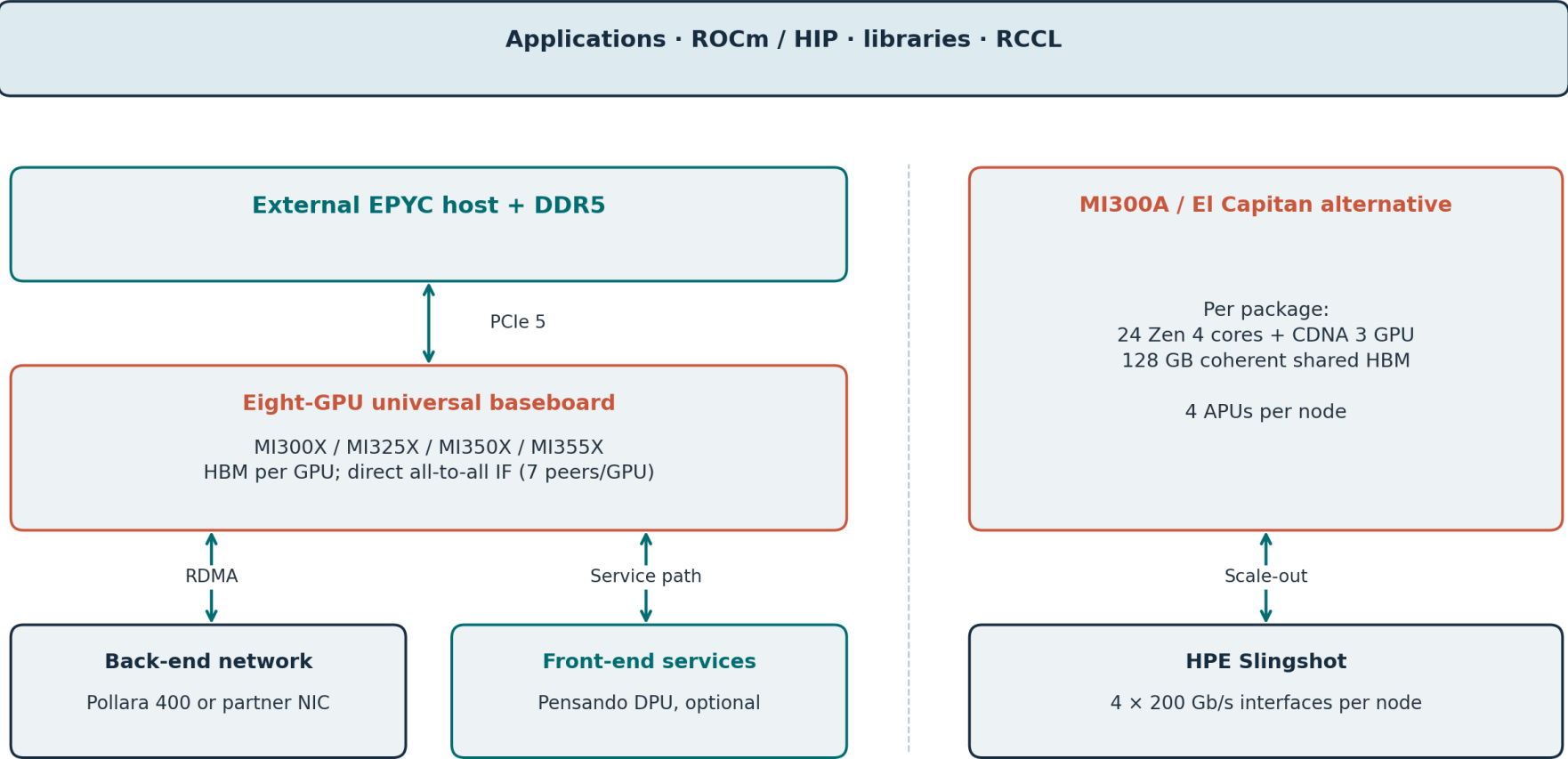


Sources: D7, D8, F8. HBM and network totals are per physical module or node, as labeled.

Frontier combines coherent node integration with a two-GCD accelerator. ROCm and application placement must account for eight logical GPUs across four physical packages. The Slingshot fabric is the scale-out network; it is not another name for the package-interior link.

2023-2025 | Two different integration choices

Discrete eight-GPU baseboards and coherent CPU-GPU APUs share an integration lineage

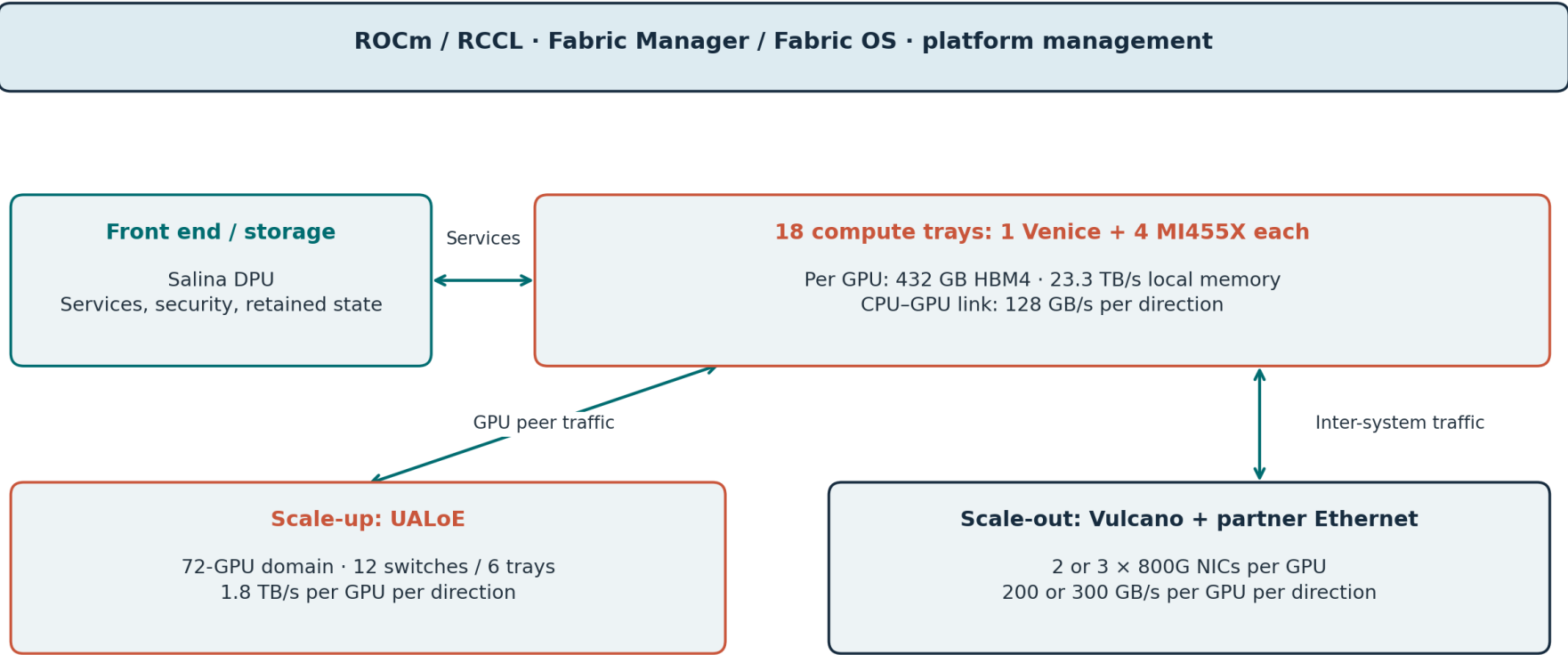


Sources: E1, E3, E7, E10, I1, I2, H6. An eight-GPU baseboard domain does not expand just because more boards occupy a rack.

MI300 splits into two system patterns. MI300A puts CPU and GPU access inside one coherent HBM package. MI300X, MI325X and MI350 use external hosts and a direct eight-GPU baseboard. Pollara can provide the back-end network, while a Pensando DPU can service the front end. These choices are complementary rather than one unavoidable component sequence.

2026 | Helios separates three communication roles

Published reference: 18 Venice CPUs + 72 MI455X GPUs; launch/production disclosed, customer rollout varies



Rack layer: liquid cooling · power delivery · serviceability · OCP-aligned design

Switch silicon, optics and manufacturing include partner components. Rack HBM is distributed, not uniformly local.

Sources: C8, E17, I4, I5, K2, K3, N14, N15. Rates are normalized from vendor bidirectional totals where applicable.

Helios combines eighteen Venice CPUs, seventy-two MI455X GPUs, a switched UALoE scale-up domain, and Vulcano scale-out attachment. The maximum published scale-out configuration uses three 800G NICs per GPU, or 300 GB/s per direction; two-NIC configurations provide 200 GB/s. Salina occupies the infrastructure-service path. Rack management, liquid cooling and ROCm/fabric software complete the design.

9.2 Bandwidth hierarchy by era

Era / configuration	On-die or package interior	Local memory per accelerator	CPU-accelerator	Scale-up per accelerator, one direction	Scale-out per accelerator, one direction	Sources
2017: EPYC + MI25	Monolithic GPU; L2 4 MB; internal data rate n/d	HBM2 484 GB/s	PCIe 3 x16: 15.75 GB/s encoding ceiling	No dedicated peer IF in this card	OEM-dependent; n/d	D1 D2 E19
2021-2022: Frontier / MI250X	2 GCDs: 200 GB/s aggregate each direction between them	3200 GB/s/package; 1600 per GCD (vendor rounded)	Coherent IF in Frontier; exact dedicated CPU path not inferred from aggregate	50 GB/s per external link; 8 links/package cover node connections	25 GB/s per physical package from node network total	D7 D8 F8
2023-2024: MI300X UBB	4 MB L2/XCD; 256 MB Infinity Cache; cited on-package bisections use different scopes	5300 GB/s/GPU	PCIe 5 x16: ~63 GB/s encoding ceiling	448 GB/s total: 7 x64; 8-GPU domain	50 GB/s if 1 x400G NIC/GPU is configured	E1 E3 N9 I2
2023-2024: MI300A / EI Capitan	CPU/GPU share coherent package HBM through four IODs	5300 GB/s/APU	On-package coherent memory; no mandatory host-copy stage	64 GB/s per link; 4 APUs/node, link allocation system-specific	25 GB/s/APU from node network total	E1 E7 F9
2025: MI355X UBB	160 KB LDS/CU; 4 MB L2/XCD; 256 MB Infinity Cache	8000 GB/s/GPU	PCIe 5 x16: ~63 GB/s encoding ceiling	537.6 GB/s total: 7 x76.8; 8-GPU domain	50 GB/s if 1 x400G Pollara/GPU is configured	E10 N12 N24 I2
2026: MI455X / Helios	192 MB global L2; 2 fabric/cache dies; interior bandwidth n/d here	23300 GB/s/GPU	128 GB/s/GPU (256 bidirectional vendor figure)	1800 GB/s total/GPU; 72-GPU switched domain	Up to 300 GB/s/GPU: 3 x800G Vulcano	N14 N15 K3 I4

The hierarchy shows why bulk memory movement cannot be modeled with one “system bandwidth.” MI355X's 8 TB/s HBM rate is about fifteen times its one-direction scale-up total and 160 times a configured 400G NIC. MI455X's larger switched fabric improves the remote path substantially, yet local HBM is still about thirteen times the one-direction scale-up total and about seventy-eight times maximum scale-out. These are peak-rate ratios, not latency or collective-performance predictions.

9.3 Technology reuse across the stack

CPU reuse: CPU reuse is real, but the surrounding machine differs

Core generation	Server products	Relevant client/workstation connection
Zen	Naples	Ryzen 1000 and first Threadripper reuse Zeppelin-based technology
Zen+	No corresponding main EPYC generation	Client process/implementation refinement; not a missing Naples-to-Rome server generation
Zen 2	Rome	Ryzen 3000 and Threadripper 3000 illustrate compute-chiplet reuse
Zen 3	Milan / Milan-X	Ryzen 5000; 5800X3D illustrates stacked cache in a client product
Zen 4	Genoa / Genoa-X; MI300A CPU chiplets	Ryzen 7000 and Threadripper 7000; Phoenix APUs show a different integration choice
Zen 4c	Bergamo / Siena	Dense-core concepts also appear in selected mobile products
Zen 5 / Zen 5c	Turin; Sorano; EPYC 4005	Ryzen 9000 and Ryzen AI products share core-generation technology with different cache and SoC balances
Zen 6 / Zen 6c	Venice	This report anchors the generation to verified server disclosures at the cutoff

An APU can use a related CPU core with different L3 capacity, memory controllers, power limits and integrated graphics. Hot Chips presentations on Raven Ridge in 2018, Ryzen 4000 in 2020, Ryzen 6000 in 2022 and Ryzen 7040 in 2023 show these distinct SoC integration choices. The shared core does not imply a shared server platform.

GPU reuse: GPU reuse changes after Vega

Early Instinct products have direct graphics relatives: Fiji, Vega and Vega 20. From MI100 onward, CDNA is the dedicated compute branch, while RDNA serves graphics-oriented products. Shared packaging and design methods do not imply equivalent compute units.

Hot Chips' Vega disclosure, ISSCC 2020's RX 5700 implementation paper and Hot Chips 2025's RDNA 4 presentation form a useful adjacent reading line. Radeon graphics/memory chiplets use a different organization from MI300's stacked compute and I/O dies. XDNA NPUs and Xilinx adaptive devices form separate accelerator lineages.

The reuse extends to integration techniques: EFB appears in MI250X and Venice disclosures; hybrid bonding supports stacked cache and GPU compute-over-I/O designs; mature-node infrastructure dies preserve expensive leading-edge area for compute. Reuse does not imply identical implementations. V-Cache's vertical interface, MI300's compute-to-IOD links and a rack-scale UALoE connection solve different distance, power and bandwidth problems.

Source map: A1-A9, B2-B14, C1-C7, D2-D10, E1-E19, G14, H6, I1-I5, J1-J13, K2-K9; N14, N15, N24-N31. Full references appear in the bibliography; numerical cells are indexed in the companion claims ledger.

10. Engineering trends and trade-offs

10.1 Chiplet economics and architecture must be read together

Smaller dies can improve manufacturing flexibility and enable reuse, but the architectural boundary determines whether the assembly performs well. Naples, Rome, MI250X, MI300 and MI350 each chose a different boundary. Treating them all as simply “chiplet products” hides the most important decisions.

For CPUs, moving memory control into an I/O die made compute scaling more modular. For GPUs, moving from two complete GCDs to compute chiplets over infrastructure dies changed the programming and memory organization. MI400 separates additional infrastructure functions while extending communication to a rack.

10.2 More cache, more HBM and more cores solve different problems

V-Cache reduces traffic when reused data remains in cache. HBM raises bandwidth and places substantial memory close to computation. More cores or matrix engines increase execution capacity. The useful combination depends on reuse, working-set size, parallelism and numerical requirements.

This is the reason for AMD's increasingly branched portfolio. Genoa-X, Bergamo, Siena and custom HBM EPYC systems address different constraints. MI300A, MI300X and later PCIe or rack-oriented Instinct products likewise represent different system balances rather than a single universal accelerator design.

10.3 Locality persists as integration increases

A coherent system is not necessarily uniform in access cost. A socket is not necessarily one cache-sharing domain. An accelerator package is not necessarily one logical GPU. A rack's aggregate memory is not necessarily a single local pool.

The practical reading of each generation should identify four boundaries: where caches are shared, where memory is physically attached, which compute resources form a logical device, and which links carry communication between those domains. Those boundaries explain many performance differences that process node and peak throughput cannot.

10.4 Infrastructure processing and communication are different workloads

The DPU and AI NIC histories clarify why a server portfolio needs more than fast CPUs and GPUs. A DPU can remove networking, security and storage work from application processors and place policy enforcement in a separate domain. An AI NIC improves communication among accelerators. FPGA-based platforms add custom data paths where the workload justifies hardware specialization.

Their performance measures differ. Stateful services depend on connections, table scale and packet processing. Collective communication depends on message delivery, congestion and the slowest participating endpoints. A product comparison must therefore start with the work being moved, then examine the memory and communication resources needed to execute it.

10.5 The platform boundary now extends across the rack

The expansion from EPYC and Instinct into networking and adaptive compute makes the system boundary explicit. HBM, host DDR, DPU memory and NVMe are separate tiers with different access costs. Scale-up and scale-out links serve different traffic. Management software decides how shared resources are partitioned and recovered.

The engineering implication is to evaluate an AI platform as a chain of execution and data movement: ingestion, preparation, model execution, communication, retained state and delivery. Improving the component that limits that chain matters more than maximizing every isolated specification.

10.6 The best source combination changes with the question

For **a core**, start with Hot Chips and the corresponding ISSCC paper. For **a package**, add the industrial product paper and VLSI/package material. For **an HPC node**, add the exascale papers and laboratory documentation. For **a cloud service**, add the provider's current VM documentation and deployment announcement. For **AI scaling**, add communication, software and rack-level disclosures.

AMD's strongest sustained advantage in this history is the ability to reuse technology while changing the product boundary. The report's product-by-product comparisons show how that capability grew from a reusable CPU SoC into a family of CPUs, GPUs, programmable network processors, adaptive accelerators and complete AI systems.

10.7 Quantitative balance across generations

Configuration	DRAM GB/s/core	L3 MB/core	HBM B/dense BF16 FLOP	HBM B/dense FP8 FLOP	Scale-up/HBM bandwidth	Scale-out GB/s/accelerator	Dense BF16 TFLOP/s per rated W
Naples	5.33	2.00	n/a	n/a	n/a	n/a	n/a
Rome	3.20	4.00	n/a	n/a	n/a	n/a	n/a
Milan	3.20	4.00	n/a	n/a	n/a	n/a	n/a
Milan-X	3.20	12.00	n/a	n/a	n/a	n/a	n/a
Genoa	4.80	4.00	n/a	n/a	n/a	n/a	n/a
Genoa-X	4.80	12.00	n/a	n/a	n/a	n/a	n/a
Bergamo	3.60	2.00	n/a	n/a	n/a	n/a	n/a
Siena	3.60	2.00	n/a	n/a	n/a	n/a	n/a
Turin classic (6400)	4.80	4.00	n/a	n/a	n/a	n/a	n/a
Turin dense (6400)	3.20	2.00	n/a	n/a	n/a	n/a	n/a
Venice 9996 RDIMM	4.00	4.00	n/a	n/a	n/a	n/a	n/a
Venice 9996 MRDIMM	6.40	4.00	n/a	n/a	n/a	n/a	n/a
MI100	n/a	n/a	0.01331	n/a	0.112	n/d	0.308
MI250X package	n/a	n/a	0.00836	n/a	0.125	25	0.684
MI300A	n/a	n/a	0.00540	0.00270	n/d	25	1.290
MI300X	n/a	n/a	0.00405	0.00203	0.085	50	1.743
MI325X	n/a	n/a	0.00459	0.00229	0.075	50	1.307
MI350X	n/a	n/a	0.00347	0.00173	0.067	50	2.307
MI355X	n/a	n/a	0.00318	0.00159	0.067	50	1.798
MI350P	n/a	n/a	0.00348	0.00174	n/d	n/d	1.917
MI455X	n/a	n/a	0.00463	0.00116	0.077	300	n/d

HBM bytes per FLOP is bandwidth divided by dense arithmetic throughput: for GB/s and TFLOP/s, divide the numerical bandwidth by 1,000 times the numerical compute rate. It is not memory capacity divided by FLOPs. DRAM bandwidth per core uses populated channel count and maximum transfer rate. Cache per core is an average capacity ratio, not a promise of an equal private allocation.

The scale-up/HBM ratio uses one direction of total accelerator link bandwidth. For MI250X, its external-port sum is a capacity bound with node-specific allocation, not an eight-peer all-to-all guarantee. Scale-out values for MI300X, MI325X and MI350 are explicit one-400G-NIC-per-GPU examples; they are not fixed silicon specifications. Frontier and El Capitan values divide their documented node network by physical accelerator packages.

Peak BF16 divided by rated watts is a deliberately narrow ratio. It gives approximately 2.31 TFLOP/s/W for MI350X and 1.80 for MI355X. This does not establish measured energy efficiency: the devices use different operating points, power may include different stated boundaries, and workload utilization determines useful output. MI455X is left n/d because a comparable final board-power figure is not established here.

10.8 The costs behind the architectural choices

Chiplets and stacking. Smaller reusable compute dies and heterogeneous nodes can improve manufacturing economics and product coverage, but add links, package complexity, test requirements and locality. Hybrid bonding supplies dense vertical connectivity; it also couples floorplanning, thermal design and assembly yield. The V-Cache latency increment is a concrete example of capacity gained at a nonzero access cost.

Precision and reuse. Arithmetic has grown faster than the ability to stream unique operands from HBM. Low-precision formats increase peak operations per byte and can reduce model storage, but require scaling metadata, numerical validation and tuned kernels. The bytes-per-FLOP trend argues for more reuse and overlap, not for assuming every format will deliver its headline rate on every model.

Scale-up and switching. The direct eight-GPU mesh offers simple one-hop peer connectivity within a small domain. Helios increases the domain to seventy-two through switches, adding congestion, failure handling and fabric-management responsibilities. Wider links help only if the communication pattern and collective implementation can use the available paths.

CPU as execution and memory tier. EPYC remains a CPU-compute product while also hosting GPUs, feeding data and running orchestration. Venice's memory-channel growth and Verano's LPDDR roadmap highlight host-memory capacity, bandwidth and power as system-design variables. Offloading inactive inference state to host or storage tiers saves HBM only when retrieval latency and bandwidth remain acceptable.

Power and supply. Higher GPU density makes liquid cooling and rack power delivery part of performance engineering. HBM stacks and advanced packaging also become production dependencies. Public product sources establish the component requirements, but do not quantify AMD's allocation, packaging yields or future supply headroom; those are not estimated here. The architectural risk is exposure to those dependencies, not an asserted production shortage.

Software and standards. Open interfaces can widen the partner ecosystem, while integration and optimization work remains substantial. A standards-compatible port does not by itself provide a stable collective stack, fleet management or efficient kernels. The useful outcome is an application that runs reliably at the intended scale; that requires the software chronology to be read alongside the hardware tables.

Source map: A7, A10, B4, C3, C8, D5, D7, D8, E1, E3, E10, E15, F1, F2, F8–F12, K2–K9; N1, N2, N5–N19, N22–N31. Full references appear in the bibliography; numerical cells are indexed in the companion claims ledger.

11. Conference record and research context

11.1 ISCA: product architecture and the exascale design history

ISCA contains the most direct bridges between AMD research and these commercial product families. The 2021 EPYC/Ryzen industrial product paper explains the evolution from reusable complete SoCs to heterogeneous compute/I/O chiplets. It is the best conceptual companion to the Naples and Rome disclosures because it explains why the partition changed.

The 2023 exascale retrospective traces the research and co-design process leading to Frontier. It treats the machine as a node and system problem, rather than simply a GPU specification. The 2024 heterogeneous-processor paper then explains MI300A and how advances in integration made the original CPU-GPU concept practical.

These three papers should be read alongside Hot Chips and ISSCC. They provide design rationale and history; the other venues provide the contemporary block diagrams and implementation results. ISCA's tutorials also matter: AMD's gem5 APU tutorial in 2018 made an ISA-level CPU/GPU research model available to the architecture community.

11.2 MICRO: interconnect, front-end efficiency and GPU translation

The 2015 paper Enabling Interposer-based Disintegration of Multi-core Processors studies how splitting a processor into smaller dies affects manufacturing cost and the on-chip/interposer network. Its importance is conceptual: partitioning silicon introduces communication costs that must be addressed architecturally. It is research on the design space, rather than documentation of Naples' actual organic-substrate implementation.

The 2020 AMD Research paper on micro-operation caches investigates fragmentation and proposes ways to use those caches more effectively. It helps explain why front-end structures are important in the Zen family, but the paper's proposed mechanisms should not be assigned to a particular Zen generation without a product disclosure.

The 2021 GPU translation paper addresses another scaling constraint: large memory capacity is useful only if address translation can reach the working set efficiently. It explores using underutilized instruction-cache and local-data-share resources to extend translation reach. The connection to Instinct is the problem being solved, not a claim that every proposed structure ships in MI200 or MI300.

MICRO also includes independent research on AMD processors and GPUs. Such measurements can reveal behavior absent from vendor diagrams. They should be distinguished from AMD-authored product disclosures and from the similarly named IEEE Micro journal.

11.3 HPCA: the heterogeneous processor before it became MI300A

HPCA 2017's Design and Analysis of an APU for Exascale Computing proposed an integrated CPU/GPU processor using chiplets and in-package stacked memory. Its evaluation considered compute resources, memory bandwidth, communication and power together. The paper's projected capacities and organization describe the research design of that period; they are not specifications for the later MI300A.

The durable contribution is the design question: which mixture of CPU execution, GPU throughput and nearby memory gives useful application performance within a system power budget? MI300A eventually embodies that general direction, but the intervening Frontier design and later packaging advances explain the differences.

HPCA 2018's GPU-modeling paper, *Lost in Abstraction*, shows why analysis at an intermediate-language level can miss behavior that appears when real machine instructions and execution resources are modeled. This matters when interpreting GPU research as evidence about commercial GCN/CDNA products.

Other AMD-affiliated HPCA work covers chiplet network design and GPU caches. These papers broaden the engineering picture: core and matrix counts are only part of a system whose behavior depends on routing, buffering, caching, translation and workload structure.

11.4 ASPLOS: extracting performance from the hardware/software boundary

ASPLOS is particularly useful for connecting the hardware roadmap to software behavior. The 2024 T3 paper proposes fine-grained overlap of computation and collective communication, using hardware/software cooperation to reduce serialized communication and contention. This is directly relevant to understanding why larger accelerator systems require more than faster links. It remains a research proposal, not a documented MI300 or Helios feature.

The 2025 recommendation-inference paper introduces the Balance scheduler, which accounts for memory-level parallelism, load imbalance and CPU-chiplet organization. Its relevance to EPYC is practical: additional cores and memory channels deliver their value only when tasks are distributed in a way that uses them effectively.

Independent ASPLOS 2024 work on explainable port-mapping inference studies AMD Zen processors through sparse performance counters. It complements vendor disclosures by helping characterize the execution resources seen by machine instructions. This is useful for compiler models and performance analysis, while remaining a different kind of source from AMD's own architecture presentations.

Together, the big four conferences show three layers of the product story: architectural ideas, engineering decisions that entered products, and the software behavior of the resulting systems. Reading across those layers connects the evolution of the hardware to the applications it ultimately serves.

11.5 Networking and adaptive computing extend the academic record

For Pensando, the direct systems bridge is NSDI 2023's joint Microsoft-AMD paper on Sirius. Its contribution is placement and state management: a shared pool of programmable cards can decouple infrastructure-service capacity from tenant-server capacity. This complements the device architecture at Hot Chips.

The broad SmartNIC literature also appears at the big four conferences, but a paper that cites Pensando is not automatically a disclosure of a Pensando product. The similarly named IEEE Micro journal published Michael Galles and Francis Matus's Pensando Distributed Services Architecture in 2021; that is the journal companion to the Hot Chips work, not a MICRO conference paper.

For the adaptive portfolio, Hot Chips 2019 and ISSCC 2020 provide a direct architecture/implementation pair for Versal. FPGA, FCCM, FPL and FPT add the mapping, compiler and hardware-design perspective. These venues matter because programming a spatial data path raises different questions from scheduling GPU threads.

11.6 Hot Chips and ISSCC: the most useful product pairings

Product or technology	Hot Chips architectural disclosure	ISSCC implementation disclosure
Zen / Naples	Zen core, 2016; EPYC architecture, 2017	Zen core, 2017; Zeppelin SoC, 2018
Vega / MI25 ancestry	Vega architecture, 2017	Use product/Vega material; no specific MI25 ISSCC paper is assumed here
Zen 2 / Rome	Zen 2, 2019	Zen 2 core and chiplet architecture, 2020
Zen 3 / Milan	Zen 3, 2021	Zen 3 core, 2022
3D V-Cache / Milan-X	Advanced-packaging tutorial, 2021	Hybrid-bonded 64 MB stacked cache, 2022
MI200	MI200 accelerator and node architectures, 2022	VLSI 2023 supplies a complementary packaging disclosure
Zen 4 / Genoa	Zen 4 and EPYC 9004, 2023	Zen 4, 2023
Zen 4c / Bergamo	EPYC 9004 family discussion, 2023	Area-optimized Zen 4c core, 2024
MI300	MI300X architecture, 2024	MI300 modular chiplet package, 2024
Zen 5 / Turin	Zen 5, 2024	Zen 5 core, 2025
MI350 / CDNA 4	CDNA 4 / MI350 architecture and platforms, 2025	MI350 stacked XCD/IOD implementation, 2026
MI400 / CDNA 5	GPU architecture and system architecture talks, 2026	No MI400 ISSCC implementation paper is established in this review
Pensando Capri / Elba	Distributed Services Architecture, 2020	Product briefs and the IEEE Micro 2021 article provide implementation context; no specific ISSCC pairing identified
Pensando Pollara 400	AI NIC architecture and application, 2025	Product and software documentation supply the device detail; no specific ISSCC pairing identified
Vulcano / Helios networking	MI400 system-architecture session, 2026	Product briefs and rack disclosures establish the deployed organization
Versal / AI Engine	VC1902 and AI Engine, 2019	7 nm ACAP processor, 2020, paper 8.2
Adaptive SmartNIC SoC	AMD 400G design, 2022	Use its disclosed interfaces and architecture without assigning a Pensando product identity

A conference year is a disclosure date, not necessarily a product introduction date. MI300's principal Hot Chips presentation was in 2024, after the 2023 launch. MI350's ISSCC paper was in 2026, after the 2025 product introduction. These distinctions prevent the conference chronology from distorting the product chronology. The DPU and NIC record has fewer direct circuit-paper pairings than Zen or Instinct; its complementary evidence is in device briefs, programming documentation and networking-systems papers.

11.7 The wider technical venues

Venue group	Contribution to understanding AMD products	Concrete examples or use in this review
ISCA, MICRO, HPCA, ASPLOS	Architecture, research rationale, measurements and software interaction	Product papers and selected research mapped in this chapter
Hot Chips	Core, SoC, accelerator and system organization	Zen/EPYC, Instinct, Pensando DSA, Pollara, adaptive SmartNIC and Versal talks
ISSCC	Circuits, physical design, SRAM, power and die integration	Zen, Zeppelin, V-Cache, MI300/MI350 and the Versal ACAP implementation
Symposium on VLSI Technology and Circuits	Packaging and architecture/implementation co-design	MI250X EFB, 2023; MI300X packaging co-optimization, 2024
IEDM	Device, process and heterogeneous-integration direction	AMD multichip keynote, 2017; chiplet/cache coevolution, 2024
DAC	Design methods, EDA and engineering strategy	Papermaster keynote, 2022; Blott keynote, 2025; AMD-affiliated integration research
DATE	Design and system-integration methodology	Chiplet opportunities/limits, 2021; next era of chiplet innovation, 2023
ISPASS and IISWC	Measurement, workloads and performance interpretation	ISPASS 2022 Rome memory characterization
SC and ISC	Complete HPC systems, applications and deployments	Frontier and El Capitan milestones; EPYC/Instinct ecosystem announcements
SCA/HPCAsia	HPC architecture and deployment co-design	Azure HBv5 paper, 2026
PACT, ICS, HPDC, CGO and PPOPP	Parallel execution, compilers, runtimes and heterogeneous systems	Relevant complementary literature; not separately audited as full proceedings here
ICCAD, ECTC and packaging workshops	Design automation, integration, test and manufacturability	Useful neighboring venues; not a substitute for verified product-specific disclosures
NSDI, SIGCOMM, OSDI and networking workshops	Network services, transport, state and operational systems	NSDI 2023 Sirius is a direct Pensando case; other venues supply complementary research
FPGA, FCCM, FPL and FPT	Reconfigurable architecture, compilation and workload mapping	Relevant to Alveo, Versal and FPGA-based network acceleration
SIGGRAPH, GDC and GPU developer events	Graphics architecture, rendering and software	Useful for Radeon/RDNA and shared technology, less direct for Instinct server architecture

DAC deserves inclusion but plays a different role from Hot Chips or ISSCC. AMD's 2022 keynote on AI and HPC in EDA, its 2021 cross-disciplinary innovation talk and its 2025 AMD Research keynote show the design-methodology context. They should not be treated as a complete EPYC or Instinct microarchitecture disclosure.

11.8 Indexed disclosures and events

The index links product architecture, implementation and deployment evidence. A paper's presence is not evidence that all of its proposed mechanisms shipped. Industry-product papers are distinguished from related research, and the IEEE Micro journal is distinct from the MICRO conference. Some archive entries group several talks or years; these are discovery guides rather than claims of a single paper with that title.

For paywalled records, this report uses the accessible abstract, program, author manuscript or vendor slides named in the bibliography. It does not claim to have reconstructed a circuit from a title alone. In particular, the ISSCC 2026 program establishes the MI350 paper and die-node pairing; the detailed architecture here comes from the available CDNA 4 documentation and product material.

Venue	Year	Paper or talk	Presenters (if known)	Product/generation	Disclosure type (architecture / circuit and implementation / packaging / system / related research)	Link
Hot Chips	2016	A New, High Performance x86 Core Design from AMD	Michael Clark	Zen / Naples	architecture	A1
Hot Chips	2017	Next Generation AMD Enterprise Server Product Architecture	Kevin Lepak	EPYC 7001 / Naples	architecture	A2
Hot Chips	2017	Radeon Vega architecture	Mike Mantor; Ben Sander	Vega / MI25 ancestry	architecture	D2
Hot Chips	2019	Zen 2	AMD	Rome; Ryzen chiplets	architecture	A5
Hot Chips	2019	Versal and AI Engine architecture	Sagheer Ahmad; Sridhar Subramanian	Versal VC1902	architecture	J5
Hot Chips	2020	Pensando Distributed Services Architecture	Francis Matus	Capri / Elba	architecture	H2
Hot Chips	2020	Versal Premium	Xilinx	Versal Premium	architecture	J7
Hot Chips	2021	Zen 3 architecture	Mark Evers	Milan	architecture	B1
Hot Chips	2021	Advanced packaging tutorial	Raja Swaminathan	Chiplets / V-Cache	packaging	B5
Hot Chips	2021	Versal AI Edge	Xilinx	Versal AI Edge	architecture	J7
Hot Chips	2022	MI200 accelerator and node architectures	Alan Smith; Norman James	MI250X / Frontier	architecture / system	D8
Hot Chips	2022	400G Adaptive SmartNIC SoC	AMD	Adaptive SmartNIC research/product SoC	architecture	I6
Hot Chips	2023	Zen 4 EPYC family processor architecture	Kai Troester; Ravi Bhargava	Genoa / Bergamo / Siena	architecture	B8
Hot Chips	2024	Next Generation Zen 5 Core	Brad Cohen; Mahesh Subramony	Turin and shared client cores	architecture	C1
Hot Chips	2024	MI300X architecture	Alan Smith; Vamsi Alla	MI300X	architecture / packaging	E3
Hot Chips	2025	CDNA 4 / MI350 architecture and platforms	Michael Floyd; Michael Steffen	MI350X / MI355X	architecture / system	E11
Hot Chips	2025	Pollara 400 AI NIC Architecture and Application	Kevin Chu	Pollara 400	architecture	I2
Hot Chips	2026	MI400 GPU architecture	Alan Smith; Maiyuran Subramaniam	MI455X / CDNA 5	architecture	E18
Hot Chips	2026	MI400 system architecture	Steve Scott; David Riddoch; Krishna Doddapaneni	Helios	system	E18
ISSCC / JSSC	2017	Zen core implementation; expanded JSSC treatment	Teja Singh et al.	Zen	circuit and implementation	A3
ISSCC	2018	Zeppelin: An SoC for Multichip Architectures, 2.4	Noah Beck et al.	Naples / Ryzen / Threadripper	circuit and implementation	A4
ISSCC	2020	Zen 2 core and chiplet architecture, 2.1 / 2.2	AMD	Rome / Ryzen	circuit and implementation / packaging	A6
ISSCC	2020	A Versatile 7nm Adaptive Compute Acceleration Platform Processor, 8.2	Prasun K. Raha et al.	Versal	circuit and implementation	J6
ISSCC	2022	Zen 3: AMD 2nd-Generation 7nm x86-64 Microprocessor Core, 2.7	Thomas Burd et al.	Milan / Ryzen 5000	circuit and implementation	B2
ISSCC	2022	3D V-Cache: Hybrid-Bonded 64MB Stacked Cache, 26.4	John Wuu et al.	Milan-X / Ryzen X3D	packaging	B4

Venue	Year	Paper or talk	Presenters (if known)	Product/generation	Disclosure type (architecture / circuit and implementation / packaging / system / related research)	Link
ISSCC	2023	Zen 4: AMD 5nm 5.7GHz x86-64 Microprocessor Core	Benjamin Munger et al.	Zen 4 core implementation	circuit and implementation	B9
ISSCC	2024	Zen 4c: AMD 5nm Area-Optimized x86-64 Microprocessor Core, 2.2	Thomas Burd et al.	Bergamo / Siena	circuit and implementation	B10
ISSCC	2024	MI300 Series Modular Chiplet Package, 11.1	AMD	MI300A / MI300X	packaging	E2
ISSCC	2025	Zen 5: AMD High-Performance 4nm x86-64 Microprocessor Core, 2.1	AMD	Zen 5 / Turin	circuit and implementation	C2
ISSCC	2026	MI350 GPUs: CDNA 4-Based 3D-Stacked 3nm XCDs and 6nm IODs, 2.1	Ramasamy Adaikkalavan et al.	MI350	circuit and implementation / packaging	E12
ISCA	2018	Modular Routing Design for Chiplet-based Systems	Jun Yin et al.	Chiplet networks; no SKU attribution	related research	F4
ISCA	2018	gem5 APU tutorial	AMD	GPU/CPU simulation	related research	F13
ISCA	2021	Pioneering Chiplet Technology and Design for EPYC and Ryzen	Samuel Naffziger et al.	Naples / Rome / Ryzen	architecture / packaging	A7
ISCA	2023	A Research Retrospective on AMD's Exascale Computing Journey	Gabriel Loh et al.	Frontier / exascale co-design	system	F8
ISCA	2024	Realizing the AMD Exascale Heterogeneous Processor Vision	Alan Smith et al.	MI300A	architecture / packaging	F9
ISCA	2026	SPEC CPU: The Next Generation	Mahesh Madhav et al.; AMD coauthors	Cross-industry benchmark methodology	related research	N53
ISCA	2026	PowerGrad: Hierarchical Power Management for Power-Limited ML Inference Clusters	Hyoungwook Nam et al.; Aporva Amarnath (AMD)	Power-management research; no SKU attribution	related research	N53
ISCA	2026	Lit Silicon: A Case Where Thermal Imbalance Couples Concurrent Execution in Multiple GPUs	Marco Kurzynski; Shaizeen Aga; Di Wu	Thermal-coupling research; program record	related research	N53
HPCA	2026	Architecting Resilience at Scale: From Research to Practice	Sudhanva Gurusurthi (AMD)	Resilience keynote; no new SKU implied	related research	N54
MICRO	2015	Enabling Interposer-based Disintegration of Multi-core Processors	Ajaykumar Kannan; Natalie Enright Jerger; Gabriel Loh	Chiplet design space	related research	F1
MICRO	2020	Improving the Utilization of Micro-operation Caches in x86 Processors	Jagadish Kotra; John Kalamatianos	Zen-relevant research; no SKU attribution	related research	F6
MICRO	2021	Increasing GPU Translation Reach by Leveraging Under-Utilized On-Chip Resources	Jagadish Kotra et al.	GPU translation; no SKU attribution	related research	F7
HPCA	2017	Design and Analysis of an APU for Exascale Computing	Akshay Vijayaraghavan et al.	Research predecessor to integrated APU	related research	F2
HPCA	2018	Lost in Abstraction: Pitfalls of Analyzing GPUs at the Intermediate Language Level	Anthony Gutierrez et al.	GPU simulation methodology	related research	F3
HPCA	2020	Experiences with ML-Driven Design: A NoC Case Study	Jun Yin et al.	Network design exploration	related research	F5
ASPLOS	2024	T3: Transparent Tracking & Triggering for Fine-grained Overlap of Compute & Collectives	Suchita Pati et al.	Compute/collective overlap; no SKU attribution	related research	F10
ASPLOS	2024	Explainable Port Mapping Inference with Sparse Performance Counters for AMD Zen	Fabian Ritter; Sebastian Hack	Independent Zen characterization	related research	F11
ASPLOS	2025	Load and MLP-Aware Thread Orchestration for Recommendation Systems Inference on CPUs	Rishabh Jain et al.	Chiplet CPU workload scheduling	related research	F12
VLSI	2023	MI250X Enabled by Elevated Fanout Bridge Advanced Packaging Architecture	Raja Swaminathan et al.	MI250X	packaging	D10
VLSI	2024	MI300X packaging and architecture co-optimization	Alan Smith et al.	MI300X	packaging	E4

Venue	Year	Paper or talk	Presenters (if known)	Product/generation	Disclosure type (architecture / circuit and implementation / packaging / system / related research)	Link
IEDM	2024	Coevolution of Chiplet Technology and Cache Architecture for AI and Compute	John Wu et al.	CPU/GPU integration	packaging	F15
ISPASS	2022	Memory Performance of AMD EPYC Rome and Intel Cascade Lake SP	Velten et al.	Independent Rome measurements	related research	A10
NSDI	2023	Disaggregating Stateful Network Functions	Deepak Bansal et al.	Sirius / Pensando DSC	system	H10
SCA/HPCAsia	2026	Cloud-Hardware Co-Design for Memory Bandwidth-Bound HPC: Azure HBv5	Microsoft/AMD collaboration	Custom EPYC with HBM	system	C11
DAC / DATE	2019-2025	Chiplet-integration research and design-methodology keynotes	AMD-affiliated authors; Mark Papermaster; Michaela Blott	Cross-portfolio; archive-level leads	related research	F14; G15
AMD FAD	2017 / 2022	EPYC branding; CPU/GPU/adaptive strategy	AMD	EPYC / CDNA roadmap	architecture / system	G1; G2
Next Horizon	2018	Rome preview and MI50/MI60 launch	AMD	Rome / Vega 20	architecture / system	D4
AMD Data Center Premiere	2021	MI200 launch and Milan-X direction	AMD	MI200 / Milan-X	architecture / system	G3
AMD Data Center and AI Premiere	2023	Bergamo, Genoa-X and MI300X preview	AMD	EPYC 9004 / MI300X	architecture / system	B13
Advancing AI	2023	MI300 launch	AMD	MI300A/X / ROCm 6	architecture / system	E6; N36
Advancing AI	2024	Turin, MI325X, Salina and Pollara introduction	AMD	CPU / accelerator / network	architecture / system	C4; E9
Advancing AI	2025	MI350 and open ecosystem; Helios preview	AMD	MI350 / Helios roadmap	architecture / system	E13
Advancing AI	2026	Full-stack compute for agentic AI	AMD and partners	Venice / MI455X / Helios / Vulcano	architecture / system	C8
Computex	2021 / 2024 / 2025	V-Cache demo; Zen 5 and Instinct roadmap; Radeon AI PRO	AMD	Shared technology and product branches	architecture / system	B7; G4; J12
CES	2026	AI everywhere and Helios preview	AMD and partners	MI400 and client/edge platforms	system	G5
OCP Global Summit	2025	Helios and Open Rack for AI	AMD / Meta ecosystem	Helios mechanical/electrical integration	system	G6
MWC	2024	Telco and edge demonstrations	AMD and partners	Siena / adaptive edge	system	G7
AWS re:Invent	2021 / 2023	M6a launch; workload deployment session	AWS / AMD	Milan and EPYC cloud systems	system	G8; G9
Microsoft Ignite / Build	2023 / 2024	MI300X preview and GA; HBv5 announcement	Microsoft / AMD	MI300X / custom EPYC	system	G10; G11; G12
Google Cloud Next	2025	C4D and H4D	Google / AMD	Turin cloud deployment	system	G13
NVIDIA GTC	2020	DGX A100 introduction	NVIDIA	EPYC 7742 as accelerator host	system	G14
SC / ISC	2020-2024	MI100 launch period; Frontier and El Capitan milestones	AMD / ORNL / LLNL / HPE	Complete HPC systems	system	D6; F8; E8
VMware Explore	2022	vSphere 8 and Distributed Services Engine	VMware / AMD	Pensando DSC	system	H11
Cisco Live	2025	N9300 Smart Switches	Cisco	Pensando-enabled switch services	system	H13
Oracle cloud events	2025	AI and converged-networking partnership	Oracle / AMD	Pensando and cloud infrastructure	system	H14

11.9 Venue coverage and limits

The strongest direct architectural record is at Hot Chips; ISSCC and VLSI supply complementary implementation and integration evidence. ISCA provides several direct industrial accounts and the exascale retrospective. MICRO, HPCA and ASPLOS add research, modeling and hardware/software studies whose relationship to products must be stated explicitly.

Company events establish launches and roadmaps. Computex and CES connect server technology to client and local AI. OCP addresses rack integration; MWC emphasizes telco and edge constraints. re:Invent, Ignite, Build and Google Cloud Next establish cloud offerings. GTC belongs here through EPYC host deployments such as DGX A100. A product announced near one of these events is not automatically presented there, and an AMD-sponsored session is not automatically a new architecture disclosure.

The 2026 academic check also identifies AMD participation in ISCA work on the next SPEC CPU suite, cluster power management and thermal coupling, plus an AMD resilience keynote at HPCA. These index entries are program-level research context; no undisclosed Turin, MI350 or MI455X mechanism is inferred from their titles.

The survey covers the material product disclosures found in these venues, with selected neighboring work at NSDI, ISPASS, SCA/HPCAAsia, IEDM and DAC/DATE. It does not assert an exhaustive search of every PACT, ICS, HPDC, CGO, PPOPP, ICCAD, ECTC, FPGA, FCCM, FPL, FPT, SIGCOMM or OFC proceeding. Those remain relevant venues for follow-up on specific compiler, packaging, transport or adaptive-design questions.

Source map: A1-A10, B1, B2, B4, B5, B7-B10, C1, C2, C11, D2, D8, D10, E2-E5, E8, E11, E12, E18, F1-F15, G1-G18, H2, H3, H10-H14, I2, I6, J5-J7; N53, N54. Full references appear in the bibliography; numerical cells are indexed in the companion claims ledger.

12. Codename crosswalk

The crosswalk separates die names, architecture names, product numbers and systems. Future names carry the roadmap disclosed by July 2026; their listed role does not establish unpublished specifications. gfx targets are software ISA identities rather than marketing generations.

Codename	Marketing family / example product	Architecture / role	Sources
Zeppelin / Naples	EPYC 7001; 7601	Zen; die name / server platform	A2 A4
Rome	EPYC 7002; 7742 / 7H12	Zen 2	A8 A9
Milan / Milan-X	EPYC 7003; 7763 / 7773X	Zen 3 / stacked-cache branch	B3 B6
Genoa / Genoa-X	EPYC 9004; 9654 / 9684X	Zen 4 / stacked-cache branch	B11 B12 B13
Bergamo	EPYC 97x4; 9754	Zen 4c cloud density	B8 B13
Siena	EPYC 8004; 8534P	Zen 4c on SP6	B14
Turin / Turin Dense	EPYC 9005; 9755 / 9965	Zen 5 / Zen 5c	C3 N1 N2
Grado	EPYC 4005; 4565P / 4585PX	Zen 5 on AM5	C9 N50
Sorano	EPYC 8005; 8635P	Zen 5-based on SP6	N39 N45
Venice	EPYC 9006; 9996 reference	Zen 6/6c family; SP7/SP8 branches	C8 N30 N31
Verano	Future EPYC / Helios 500 host	2027 LPDDR-integrated host roadmap	N22 N23 C8
Florence / Ferrara / Fidenza	Future EPYC roadmap	Zen 7 generation announced for 2028; Ferrara in Helios 600	C8
Ravenna	Future EPYC roadmap	Zen 8 roadmap, 2030	C8
Polaris / Fiji / Vega 10	Radeon Instinct MI6 / MI8 / MI25	GCN graphics-derived compute	D1 D2
Vega 20	Radeon Instinct MI50 / MI60	7nm GCN compute	D3
Arcturus	Instinct MI100	CDNA; gfx908	D5 E19
Aldebaran	Instinct MI210 / MI250 / MI250X	CDNA 2; gfx90a	D7 E19
MI300 series	Instinct MI300A / MI300X / MI325X	CDNA 3; gfx942; A adds Zen 4	E1 E19
MI350 series	Instinct MI350X / MI355X / MI350P	CDNA 4; gfx950	E10 E19
MI400 series	Instinct MI455X; future MI430X	CDNA 5; AI and HPC branches	E15 E16
Capri / Elba / Giglio / Salina	Pensando DPU generations	Infrastructure services processors	H2 H4 H5 H6
Pollara	Pensando Pollara 400	400G AI NIC	I1 I2
Vulcano	Pensando Vulcano 800	800G AI NIC	I4 I5
Como / Monza	Future Helios 500 networking	2027 roadmap; product details n/d	C8
Palma / Levanzo	Future Helios 600 networking	2028 roadmap; product details n/d	C8

Codename	Marketing family / example product	Architecture / role	Sources
Helios / Helios 500 / Helios 600	Rack platform generations	MI400 / MI500 / MI600 platforms	C8 E17
XCU26	Alveo SN1022 / U45N	16nm UltraScale+ FPGA with discrete Arm complex	I7 I8
XCUX35	Alveo X3522 / X3522PV	UltraScale+ low-latency NIC / programmable variant	N49
VC1902	Versal AI Core	7nm adaptive SoC and AI Engine array	J5 J6
Versal HBM	Alveo V80	Adaptive accelerator with HBM2e	J2 J8

Source map: A2, A4, A8, A9, B3, B6, B8, B11-B14, C3, C8, C9, D1-D3, D5, D7, E1, E10, E15-E19, H2, H4-H6, I1-I10, J2, J5, J6, J8; N1, N2, N22, N23, N30, N31, N39, N45, N49, N50. Full references appear in the bibliography; numerical cells are indexed in the companion claims ledger.

13. Bibliography and source-use guide

References are grouped by evidence type. The conference index and chapter source maps connect these entries to the narrative; the companion ledger maps numerical cells to source keys and disclosure dates. Original source identifiers are retained so the revised report remains traceable to its earlier edition.

Public technical documents and author copies were used where available. Some IEEE/ACM pages restrict full-text access and some conference archives reject automated downloads; the bibliography identifies the accessible program, abstract, author manuscript or vendor-slide version where one is used. The research-data archive includes link-check results and access limitations. A DOI or program entry is not presented as evidence that the full paper was read.

Current product pages were checked at the cutoff. Material revisions include MI325X's 288 GB preview becoming 256 GB at launch, Turin's current DDR5-6400 specification versus the earlier 6000 MT/s guide, and the unresolved process-node wording in the EPYC 8005 guide. The chronology and tables use the final product or latest consistent primary specification.

13.1 Conference papers, by venue and year

- A3.** Teja Singh et al., Zen: A Next-Generation High-Performance x86 Core, ISSCC 2017; expanded journal treatment, Zen: An Energy-Efficient High-Performance x86 Core, JSSC. [Journal DOI](#).
- A4.** Noah Beck et al., Zeppelin: An SoC for Multichip Architectures, ISSCC 2018, paper 2.4. [DOI](#); [AMD presentation](#).
- A6.** ISSCC 2020, papers 2.1 and 2.2: Zen 2 core and AMD chiplet architecture for server/desktop products. [Conference press material](#).
- J6.** Prasun K. Raha et al., A Versatile 7nm Adaptive Compute Acceleration Platform Processor, ISSCC 2020, paper 8.2, pp. 146–148. [DOI](#); [conference program](#).
- B2.** Thomas Burd et al., Zen 3: The AMD 2nd-Generation 7nm x86-64 Microprocessor Core, ISSCC 2022, paper 2.7. [AMD slides](#).
- B4.** John Wu et al., 3D V-Cache: The Implementation of a Hybrid-Bonded 64MB Stacked Cache for a 7nm x86-64 CPU, ISSCC 2022, paper 26.4. [AMD slides](#).
- B9.** Benjamin Munger et al., Zen 4: The AMD 5nm 5.7GHz x86-64 Microprocessor Core, ISSCC 2023. [Conference press material](#).
- B10.** Thomas Burd et al., Zen 4c: The AMD 5nm Area-Optimized x86-64 Microprocessor Core, ISSCC 2024, paper 2.2. [DOI](#).
- E2.** AMD, MI300 Series Modular Chiplet Package—HPC and AI Accelerator for Exa-Class Systems, ISSCC 2024, paper 11.1. [IEEE record](#).
- C2.** AMD, Zen 5: The AMD High-Performance 4nm x86-64 Microprocessor Core, ISSCC 2025, paper 2.1. [IEEE record](#).
- E12.** Ramasamy Adaikkalavan et al., MI350 stacked 3 nm XCD / 6 nm IOD implementation, ISSCC 2026, paper 2.1. [DOI](#); [conference program](#).
- A7.** Samuel Naffziger et al., Pioneering Chiplet Technology and Design for the AMD EPYC and Ryzen Processor Families, ISCA 2021, industrial product paper. [DOI](#).
- F8.** Gabriel Loh et al., A Research Retrospective on AMD's Exascale Computing Journey, ISCA 2023. [Author-hosted paper](#).
- F9.** Alan Smith et al., Realizing the AMD Exascale Heterogeneous Processor Vision, ISCA 2024, industry product paper. [Author-hosted paper](#).
- D10.** Raja Swaminathan et al., AMD Instinct MI250X Accelerator Enabled by Elevated Fanout Bridge Advanced Packaging Architecture, VLSI 2023. [Author publication index](#).
- E4.** Alan Smith et al., MI300X packaging and architecture co-optimization, VLSI 2024. [Author publication index](#).
- F15.** John Wu et al., Coevolution of Chiplet Technology and Cache Architecture for AI and Compute, IEDM 2024. [Author publication index](#).

H10. Deepak Bansal et al., Disaggregating Stateful Network Functions, NSDI 2023. [Conference record](#); [paper](#). Joint Microsoft–AMD Pensando Sirius design.

C11. Cloud-Hardware Co-Design for Memory Bandwidth-Bound HPC Workloads: Performance and Characteristics of Azure HBv5 Virtual Machines, SCA/HPCAsia 2026. [DOI](#).

13.2 Conference talks, slides and recordings

A1. Michael Clark, A New, High Performance x86 Core Design from AMD, Hot Chips 28, 2016. [Presentation](#).

A2. Kevin Lepak, Next Generation AMD Enterprise Server Product Architecture, Hot Chips 29, 2017. [Presentation](#).

A5. AMD, Zen 2, Hot Chips 31, 2019. [Presentation](#).

B1. Mark Evers, Zen 3 architecture presentation, Hot Chips 33, 2021. [Conference program](#).

B5. Raja Swaminathan, advanced packaging tutorial, Hot Chips 33, 2021. [Presentation](#).

B8. Kai Troester and Ravi Bhargava, Zen 4 and EPYC 9004, Hot Chips 2023. [Presentation](#).

C1. Brad Cohen and Mahesh Subramony, Zen 5, Hot Chips 2024. [Presentation](#).

D2. Mike Mantor and Ben Sander, Vega architecture, Hot Chips 29, 2017. [Presentation](#).

D8. Alan Smith and Norman James, MI200 accelerator and node architectures, Hot Chips 34, 2022. [Presentation](#).

E3. Alan Smith and Vamsi Alla, MI300X architecture, Hot Chips 2024. [Presentation](#).

E11. Michael Floyd and Michael Steffen, CDNA 4 / MI350 architecture and platforms, Hot Chips 2025. [Conference program](#).

E18. Alan Smith and Maiyuran Subramaniam, MI400 GPU architecture; Steve Scott, David Riddoch and Krishna Doddapaneni, MI400 system architecture, Hot Chips 2026. [Conference program](#).

G15. DAC, keynote and technical-talk archive, including AMD's 2021, 2022 and 2025 speakers. [DACTv](#).

G16. Hot Chips archives, including client APU and Radeon presentations. [Archive index](#); [2020 program](#); [2022 program](#); [2023 program](#).

G17. ISSCC past-conference archive. [Programs and press materials](#). The 2020 program includes the RX 5700 implementation paper.

H2. Francis Matus, Pensando Distributed Services Architecture, Hot Chips 32, 18 August 2020. [Presentation](#). Includes Capri/Elba process, memory, interfaces and processing organization.

I2. Kevin Chu, AMD Pensando Pollara 400 AI NIC Architecture and Application, Hot Chips 2025. [Presentation](#).

I6. AMD, 400G Adaptive SmartNIC SoC, Hot Chips 34, 23 August 2022. [Presentation](#).

J5. Sagheer Ahmad and Sridhar Subramanian, Xilinx Versal / AI Engine, Hot Chips 2019. [Presentation](#).

J7. Xilinx, Versal Premium at Hot Chips 2020 and AI Edge at Hot Chips 2021. [2020 program](#); [2021 presentation](#).

13.3 Vendor technical documentation

A9. AMD, AMD EPYC 7002 Series Processors: HPC Tuning Guide, publication 56827. [Technical guide](#).

B3. AMD, Overview of AMD EPYC 7003 Series Processors Microarchitecture. [Technical document](#).

B6. AMD, Milan-X / 3D V-Cache product brief. [Brief](#).

B11. AMD, 4th Gen AMD EPYC Processor Architecture. [White paper](#).

B15. AMD, Infinity Guard generation comparison. [Security feature overview](#).

C3. AMD, EPYC 9005 Processor Architecture Overview, publication 58462, April 2025. [Document record](#); [PDF](#).

C5. AMD, 5th Gen AMD EPYC Processor Architecture. [White paper](#).

C6. AMD, EPYC 9006 portfolio. [Product family](#); [EPYC 9996 specifications](#).

C10. AMD, EPYC 8005 overview, 19 May 2026. [Product discussion](#); [architecture document 72128](#).

- D5.** AMD, CDNA Architecture. [White paper](#).
- D7.** AMD, CDNA 2 Architecture. [White paper](#).
- E1.** AMD, CDNA 3 Architecture. [White paper](#).
- E5.** Interconnect Design for Heterogeneous Integration of Chiplets in AMD Instinct MI300X Accelerator, IEEE Micro 45(1), 2025. [Author publication index](#). This is a journal article.
- E10.** AMD, CDNA 4 Architecture. [White paper](#).
- E14.** AMD, MI350P. [Product page](#); [technical deployment discussion](#).
- E15.** AMD, CDNA 5 architecture and enabled product features. [Architecture overview](#).
- E16.** AMD, MI455X. [Product specifications](#); [MI400 family](#).
- E17.** AMD, Helios. [Rack architecture and specifications](#).
- E19.** AMD ROCm, accelerator specifications and logical architecture resources. [GPU specification table](#); [MI350 microarchitecture](#).
- E20.** AMD, accelerator product specification archive. [Product table](#).
- G18.** AMD, Zen architecture overview. [Core-generation context](#).
- H3.** Michael Galles and Francis Matus, Pensando Distributed Services Architecture, IEEE Micro 41(2), 43–49, 2021. [DOI](#). Journal companion, distinct from the MICRO conference.
- H4.** AMD, Second Generation Elba Data Processing Unit, April 2024. [Product brief](#).
- H5.** AMD, Second Generation Plus Giglio Data Processing Unit, April 2024. [Product brief](#); [2023 roadmap announcement](#).
- H6.** AMD, Salina DPU. [Product brief](#); [DPU portfolio and implementation footnotes](#).
- H7.** AMD, Pensando DPU Software, 8 September 2023. [Software architecture](#).
- H8.** AMD, Software-in-Silicon Development Kit, November 2023. [SSDK brief](#); [Hello World guide](#).
- H9.** AMD, Architecture Matters: Performance of a Multi-Stage SDN Pipeline on Arm vs AMD Pensando Programmable Silicon. [Architecture and performance report](#). Vendor comparison; its benchmark claims are not used as general performance guarantees here.
- I1.** AMD, Pollara 400 AI NIC. [Product brief](#).
- I4.** AMD, Vulcano 800 AI NIC. [Product brief](#), including adapter and four-/six-NIC board configurations.
- I5.** AMD, Vulcano 800 AI NIC: Built to Scale-Out and Across, 23 July 2026. [Architecture and deployment discussion](#); [current AI NIC portfolio](#).
- I7.** AMD, Alveo SN1000 SmartNICs. [Product brief](#); [product page](#) and [new-design guidance](#).
- I8.** AMD, Alveo U45N Network Accelerator. [Product and development resources](#); [adaptive network-acceleration overview](#).
- I9.** AMD, Solarflare/Alveo adapter lineage. [X3 product page](#); [X3 migration guide](#); [X4 migration guide](#).
- I10.** AMD, Solarflare X4 Series. [Product brief](#); [product page](#).
- J1.** AMD, Alveo portfolio. [Current product map](#); [historical platform guide](#), UG1120.
- J2.** AMD, Alveo V80 Compute Accelerator, 2024. [Product brief](#); [development and product overview](#).
- J3.** AMD, Alveo V70. [Product specifications and discontinuation notice](#); [CES 2023 introduction](#).
- J8.** AMD, Versal series organization. [Family map](#); [series comparison](#), UG1273.
- J9.** AMD, Versal Premium Gen 2. [Product architecture](#); [2024 introduction](#); [Memory on Package announcement and schedules](#), June 2026.
- J10.** AMD, EPYC Embedded. [Family and platform map](#); [9005 announcement](#), March 2025.
- J11.** AMD, Kria portfolio. [K24 introduction](#), September 2023; [Kria AI SOM and robotics platform](#), 2026.
- K2.** AMD, AI Networking Built for Scale, 23 July 2026. [Salina](#), [UALoE](#), [Vulcano](#) and [fabric software](#).

- K3.** AMD ROCm, Introducing AMD CDNA 5 and the AMD Helios Rackscale Solution, 4 August 2026. [Architecture and topology](#).
- K5.** AMD, ROCm software and SDK. [Platform overview](#); [technical documentation](#).
- K6.** AMD, ROCm 10: Bringing ROCm.AI's AI-Native Developer Experiences to AMD Platforms, 27 August 2026. [Release discussion](#).
- K7.** AMD, Enterprise AI Reference Stack. [Platform components](#); [solution-blueprint documentation](#).
- K8.** AMD, Vitis AI. [Current documentation](#); [historical deep-learning processing unit architecture](#); [NPU transition documentation](#).
- K9.** AMD, Onload software. [Supported adapter families](#).
- N1.** AMD EPYC 9755 product specifications. current 2026-09-27. [Primary source](#).
- N2.** AMD EPYC 9965 product specifications. current 2026-09-27. [Primary source](#).
- N3.** AMD Zen core architecture generation comparison. current 2026-09-27. [Primary source](#).
- N5.** AMD Instinct MI100 product specifications. 2020-11-16; current page. [Primary source](#).
- N6.** AMD Instinct MI250X product specifications. 2021-11-08; current page. [Primary source](#).
- N7.** AMD Instinct MI210 product specifications. 2022-03-22; current page. [Primary source](#).
- N8.** AMD Instinct MI300A product specifications. 2023-12-06; current page. [Primary source](#).
- N9.** AMD Instinct MI300X product specifications. 2023-12-06; current page. [Primary source](#).
- N10.** AMD Instinct MI325X product specifications. 2024-10-10; current page. [Primary source](#).
- N11.** AMD Instinct MI350X product specifications. 2025-06-12; current page. [Primary source](#).
- N12.** AMD Instinct MI355X product specifications. 2025-06-12; current page. [Primary source](#).
- N13.** AMD Instinct MI350P product specifications. 2026-07; current page. [Primary source](#).
- N14.** AMD Instinct MI455X product brochure. 2026-07. [Primary source](#).
- N15.** AMD Helios blueprint brochure. 2026-07. [Primary source](#).
- N16.** AMD Q2 2026 Form 10-Q: export licensing and acquisitions. 2026-08-05. [Primary source](#).
- N17.** ROCm 7.0 release explanation. 2025-09-16. [Primary source](#).
- N18.** ROCm consolidated changelog. current 2026-09-27. [Primary source](#).
- N19.** ROCm 7.2.2 compatibility matrix. 2026. [Primary source](#).
- N20.** AMD Secure Encrypted Virtualization generation capabilities. current 2026-09-27. [Primary source](#).
- N24.** AMD Instinct MI350 Series microarchitecture. current 2026-09-27. [Primary source](#).
- N25.** AMD Instinct MI300 Series microarchitecture. current 2026-09-27. [Primary source](#).
- N26.** AMD Instinct MI250 Series microarchitecture. current 2026-09-27. [Primary source](#).
- N27.** AMD Instinct MI100 microarchitecture. current 2026-09-27. [Primary source](#).
- N28.** AMD GPU data types and precision support. current 2026-09-27. [Primary source](#).
- N29.** AMD Instinct MI355X brochure. 2025-06. [Primary source](#).
- N30.** AMD EPYC 9006 product family. current 2026-09-27. [Primary source](#).
- N31.** AMD EPYC 9996 product specifications. 2026-07; current page. [Primary source](#).
- N32.** Server processor specification archive. current 2026-09-27. [Primary source](#).
- N33.** EPYC 7002 HPC tuning guide, document 56827. 2020. [Primary source](#).
- N37.** AMD accelerator specification archive. current 2026-09-27. [Primary source](#).
- N39.** AMD EPYC 8005 architecture overview, document 72128. 2026. [Primary source](#).
- N40.** AMD Instinct MI350P PCIe GPUs: enterprise deployment. 2026-05-07. [Primary source](#).

- N44.** Xilinx fiscal 2020 Q2 Form 10-Q: Solarflare acquisition. 2019-10. [Primary source](#).
- N45.** AMD EPYC 8635P specifications. 2026-05-19; current page. [Primary source](#).
- N46.** AMD EPYC 8005 data sheet. 2026-05. [Primary source](#).
- N47.** AMD EPYC 4005 data sheet. 2025. [Primary source](#).
- N48.** AMD EPYC 4565P specifications. 2025-05-13; current page. [Primary source](#).
- N49.** AMD Alveo X3 product brief. 2024. [Primary source](#).
- N50.** AMD EPYC processor export compliance metrics; Grado identity. 2026. [Primary source](#).

13.4 Company-event keynotes and sessions

- A8.** AMD, second-generation EPYC launch, 7 August 2019. [Announcement](#).
- B7.** AMD, Computex 2021 technology announcement. [V-Cache demonstration](#).
- B12.** AMD, Genoa launch, 10 November 2022. [Announcement](#).
- B13.** AMD, Data Center and AI Technology Premiere, 13 June 2023. [Bergamo, Genoa-X and MI300X announcement](#).
- B14.** AMD, EPYC 8004 launch, 18 September 2023. [Siena announcement](#).
- C4.** AMD, fifth-generation EPYC launch, 10 October 2024. [Announcement](#).
- C7.** AMD, Taiwan ecosystem investments and Venice packaging, 21 May 2026. [Announcement](#).
- C8.** AMD, AAI 2026: AMD Delivers Full-Stack Compute for the Agentic AI Era, 23 July 2026. [Launch and roadmap](#).
- C9.** AMD, EPYC 4005 launch, 2025. [Announcement](#).
- D1.** AMD, Radeon Instinct introduction, December 2016. [MI6/MI8/MI25 announcement](#).
- D3.** AMD, MI50/MI60 introduction, 6 November 2018. [Announcement](#).
- D4.** AMD, Next Horizon, 6 November 2018. [Event announcement](#).
- D6.** AMD, MI100 introduction, 16 November 2020. [Announcement](#); [product specifications](#).
- D9.** AMD, MI200 launch, 8 November 2021. [Announcement](#).
- E6.** AMD, MI300A/MI300X launch, 6 December 2023. [Announcement](#).
- E9.** AMD, MI325X launch, 10 October 2024. [Final product announcement](#).
- E13.** AMD, Advancing AI 2025, 12 June 2025. [MI350 and ecosystem announcement](#).
- G1.** AMD, EPYC branding at Financial Analyst Day, May 2017. [Announcement](#).
- G2.** AMD, Financial Analyst Day 2022 strategy. [Announcement](#).
- G3.** AMD, Accelerated Data Center Premiere, November 2021. [Event](#).
- G4.** AMD, Computex 2024 keynote. [Presentation](#).
- G5.** AMD, CES 2026. [Announcement](#).
- G6.** AMD, OCP 2025 Helios announcement. [Open-rack direction](#); [architecture discussion](#).
- G7.** AMD, MWC 2024 ecosystem. [Telco and edge demonstrations](#).
- H1.** AMD, completion of the Pensando acquisition, 26 May 2022. [Announcement](#).
- I3.** AMD, Pollara purchase availability, 9 April 2025. [Scale-out networking announcement](#). The October 2024 introduction is also covered by reference E9.
- J4.** AMD, Alveo MA35D, 6 April 2023. [Launch](#); [product brief](#); [media portfolio](#).
- J12.** AMD, Radeon AI PRO R9700 and workstation products, Computex 2025. [Announcement](#); [R9700 specifications](#).
- J13.** AMD, completed Xilinx acquisition, February 2022. [Company filing recording the closing dates of Xilinx and Pensando](#).
- K1.** AMD, Advancing AI 2026 portfolio launch, 23 July 2026. [EPYC, Instinct, Helios and physical-AI announcement](#).

- N4.** AMD EPYC datacenter processor launch. 2017-06-20. [Primary source](#).
- N22.** AMD Meta partnership: Venice and Verano. 2026-02-24. [Primary source](#).
- N23.** AMD EPYC Venice production ramp. 2026-05-21. [Primary source](#).
- N35.** AMD MI325X initial 288 GB preview roadmap. 2024-06-03. [Primary source](#).
- N36.** AMD ROCm 6 launch. 2023-12-06. [Primary source](#).
- N43.** AMD closes ZT Systems acquisition. 2025-03-31. [Primary source](#).
- N51.** AMD completes Silo AI acquisition. 2024-08-12 announcement. [Primary source](#).
- N52.** AMD completes Nod.ai acquisition. 2023-10. [Primary source](#).

13.5 Partner disclosures

- C12.** Microsoft, current Azure HBv5 VM specifications. [HBv5 documentation](#).
- E7.** LLNL, El Capitan hardware. [Platform overview](#); [hardware guide](#).
- E8.** LLNL, SC24 and El Capitan. [Conference account](#).
- G8.** AWS, M6a launch at re:Invent, 29 November 2021. [Milan-based instances](#).
- G9.** AWS re:Invent 2023, AMD-sponsored deployment session. [Presentation](#).
- G10.** AMD and Microsoft, Ignite 2023. [Azure CPU/GPU announcement](#).
- G11.** Microsoft, Ignite 2024 Book of News. [HBv5 and other announcements](#).
- G12.** AMD, Microsoft Build 2024. [Azure MI300X availability](#).
- G13.** AMD, Google Cloud Next, 9 April 2025. [Turin-powered C4D and H4D](#).
- G14.** NVIDIA, DGX A100 introduction at GTC 2020. [Announcement](#); [system guide](#).
- H11.** AMD, Pensando DSC and VMware vSphere 8, 30 August 2022. [Announcement](#).
- H12.** HPE, Pensando in distributed switch services. [Secure interconnect architecture](#). Current switch QuickSpecs are listed under N34.
- H13.** Cisco, N9300 Smart Switches, Cisco Live, February 2025. [Announcement](#); [current data sheet](#).
- H14.** AMD and Oracle, AI and converged-networking partnership, 2025. [Announcement](#).
- K4.** AMD and HPE, rack-scale collaboration, December 2025. [UALoE switch ecosystem](#); AMD/Broadcom, Advancing AI 2026, [Ethernet scale-up session](#).
- N21.** Sanmina completes acquisition of ZT manufacturing. 2025-10-27. [Primary source](#).
- N34.** HPE CX 10000 and CX 10040 QuickSpecs. 2026. [Primary source](#).
- N41.** Cirrascale Helios and MI400 availability plan. 2026-07. [Primary source](#).
- N42.** Microsoft to deploy next-generation AMD processors. 2026-07-20. [Primary source](#).

13.6 Related research

- F1.** Ajaykumar Kannan, Natalie Enright Jerger and Gabriel Loh, Enabling Interposer-based Disintegration of Multi-core Processors, MICRO 2015. [Author-hosted paper](#).
- F2.** Akshay Vijayaraghavan et al., Design and Analysis of an APU for Exascale Computing, HPCA 2017. [Author-hosted paper](#).
- F3.** Anthony Gutierrez et al., Lost in Abstraction: Pitfalls of Analyzing GPUs at the Intermediate Language Level, HPCA 2018. [Author-hosted paper](#). The conference year is 2018 despite the PDF filename.
- F4.** Jun Yin et al., Modular Routing Design for Chiplet-based Systems, ISCA 2018. [Author publication index](#).
- F5.** Jun Yin et al., Experiences with ML-Driven Design: A NoC Case Study, HPCA 2020, industrial track. [Author publication index](#).

F6. Jagadish Kotra and John Kalamatianos, Improving the Utilization of Micro-operation Caches in x86 Processors, MICRO 2020. [Author-hosted paper](#).

F7. Jagadish Kotra et al., Increasing GPU Translation Reach by Leveraging Under-Utilized On-Chip Resources, MICRO 2021. [Author-hosted paper](#).

F10. Suchita Pati et al., T3: Transparent Tracking & Triggering for Fine-grained Overlap of Compute & Collectives, ASPLOS 2024. [Author manuscript](#); [DOI](#).

F12. Rishabh Jain et al., Load and MLP-Aware Thread Orchestration for Recommendation Systems Inference on CPUs, ASPLOS 2025. [Author institution record](#); [conference program](#).

F13. AMD gem5 APU tutorial, ISCA 2018. [Tutorial archive](#).

F14. Gabriel Loh et al., chiplet-integration studies at DATE 2021 and DATE 2023; AMD-affiliated integration research at DAC 2019. [Author publication index](#).

N53. ISCA 2026 conference program: benchmark and power-management research. 2026-06 to 2026-07. [Primary source](#).

N54. HPCA 2026 program: Architecting Resilience at Scale. 2026. [Primary source](#).

13.7 Independent measurements

A10. Velten et al., Memory Performance of AMD EPYC Rome and Intel Cascade Lake SP Server Processors, ISPASS 2022. [Author manuscript](#).

F11. Fabian Ritter and Sebastian Hack, Explainable Port Mapping Inference with Sparse Performance Counters for AMD's Zen Architectures, ASPLOS 2024. [Author-hosted paper](#).

13.8 Source navigation for current specification checks

The N-series supplements the conference record with current CPU and GPU specification pages, ROCm release and compatibility material, the final MI455X/Helios brochures, corporate transaction records, and card-specific network documents. N1–N4 support the CPU baseline and current Turin comparison; N5–N15 support accelerator and rack tables; N16 identifies regional shipments; N17–N20 support software and security; N21–N23 and N41–N44 establish transaction, production and deployment timing; N24–N29 give accelerator programming and format context; N30–N37 support portfolio, specification and chronology checks; N39–N40 and N45–N50 support the Sorano, MI350P, AM5 and network-card corrections. N51–N52 document software acquisitions; N53–N54 check the 2026 academic program record. N38 was discarded because its proposed product URL did not resolve to a verified document.

H3 and E5 are IEEE Micro journal articles, not MICRO conference papers. K1 duplicates the Advancing AI 2026 event recorded under C8 and is retained for compatibility with the original report. Bibliography links that point to an author's publication index are explicitly discovery/record links; only the technical detail established in the reviewed materials is used.